

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

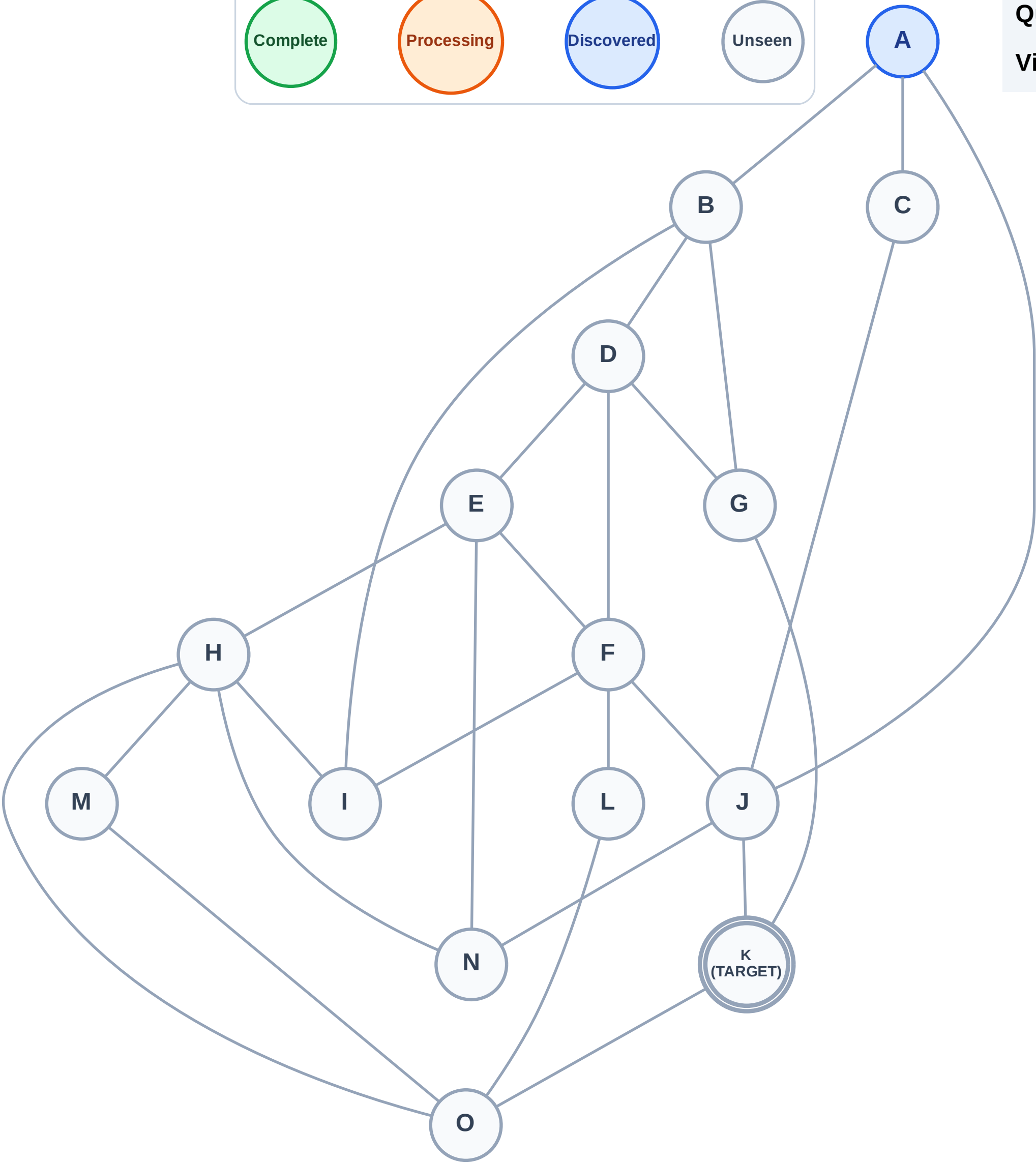
Complete

Processing

Discovered

Unseen

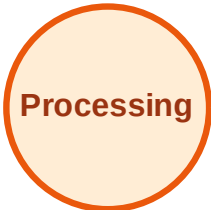
Queue: A
Visited: (none)



BFS Traversal

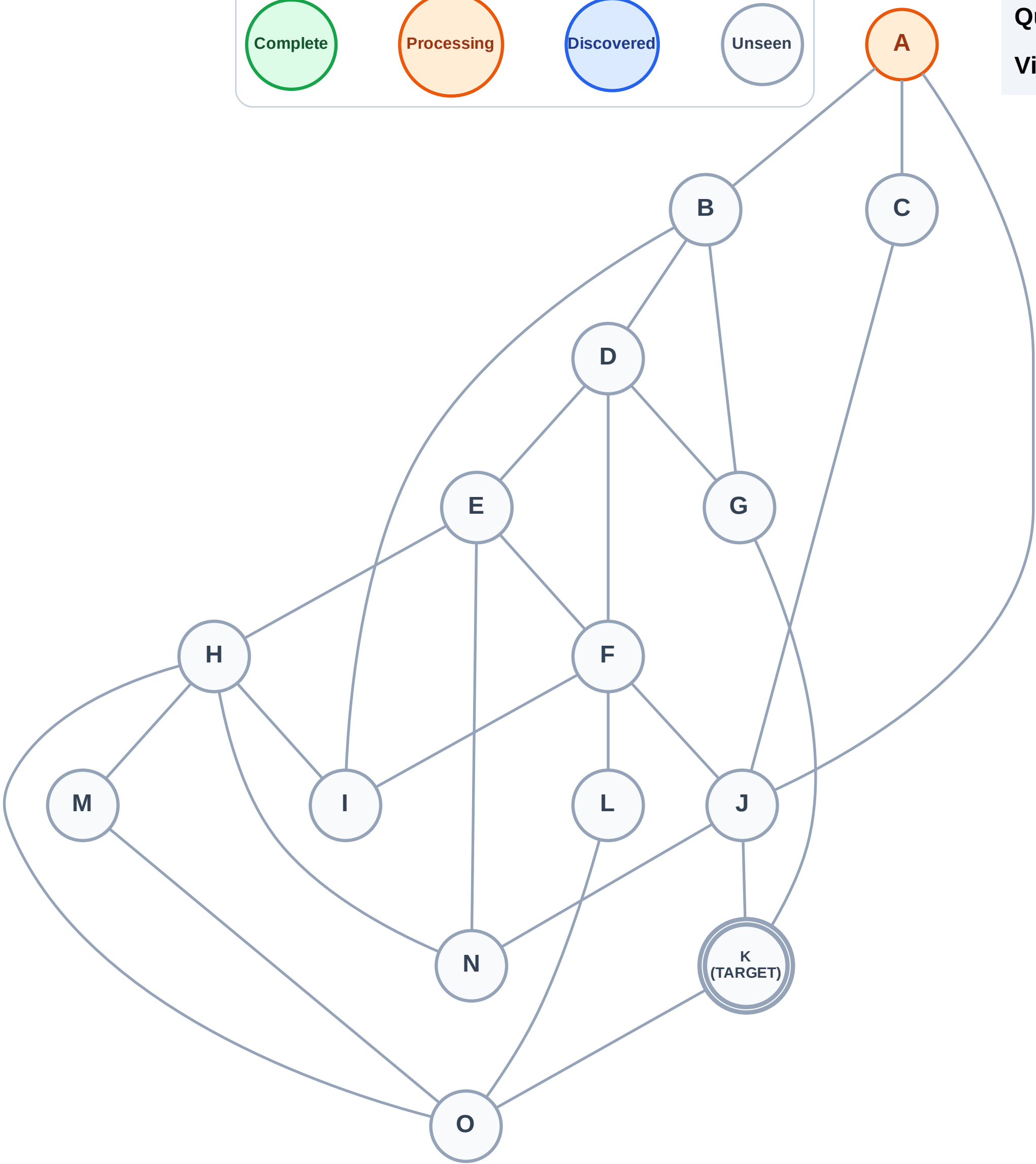
Step 2: Process node A

Legend



Queue: (empty)

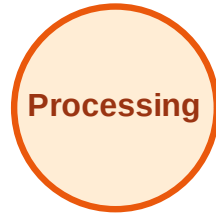
Visited: (none)



BFS Traversal

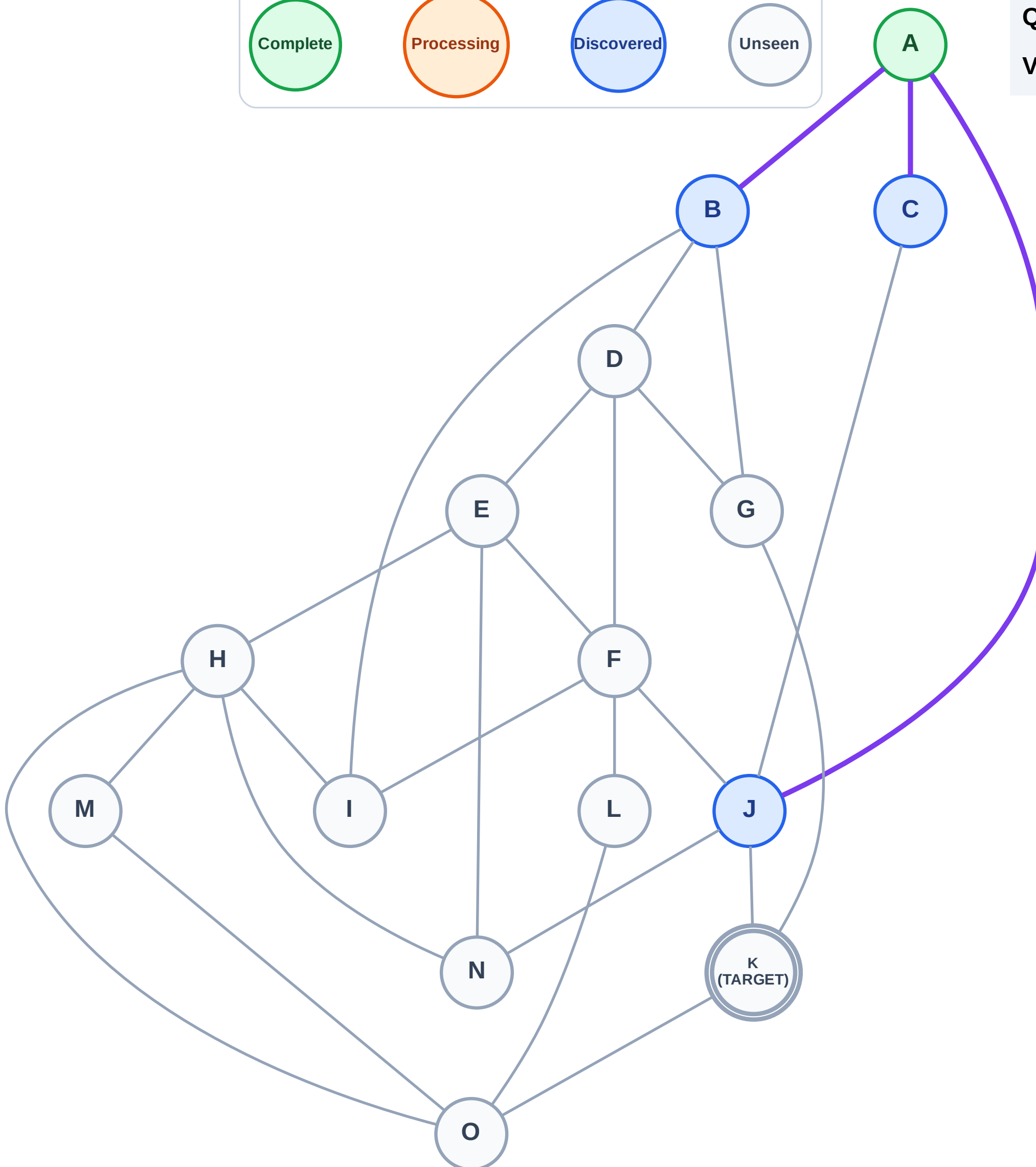
Step 3: Discover B, C, J from A

Legend



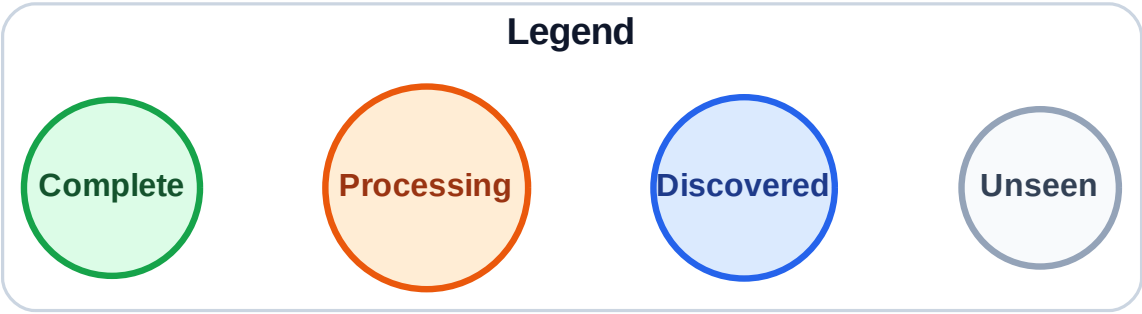
Queue: B → C → J

Visited: A

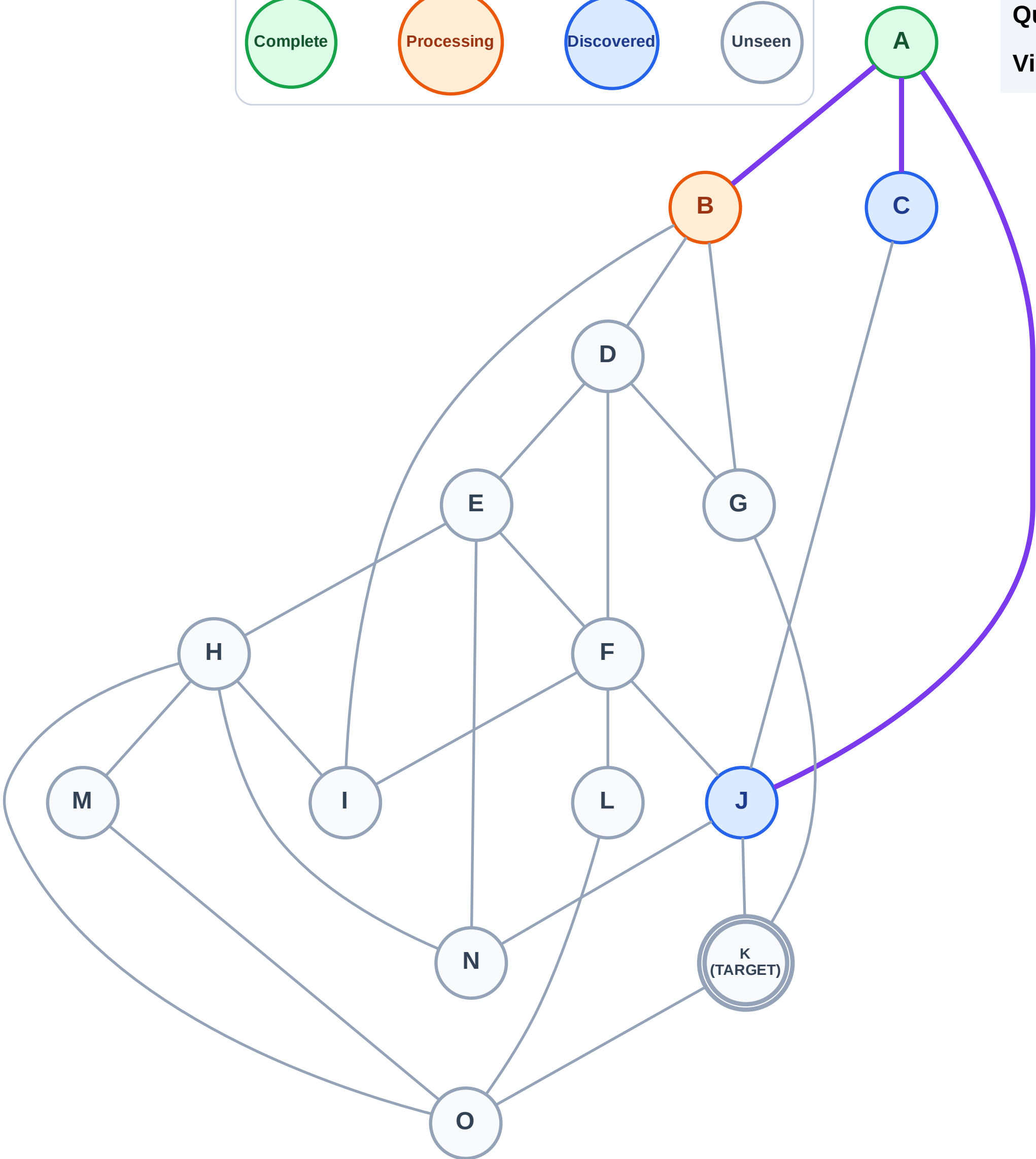


BFS Traversal

Step 4: Process node B



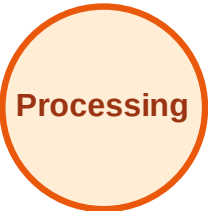
Queue: C → J
Visited: A



BFS Traversal

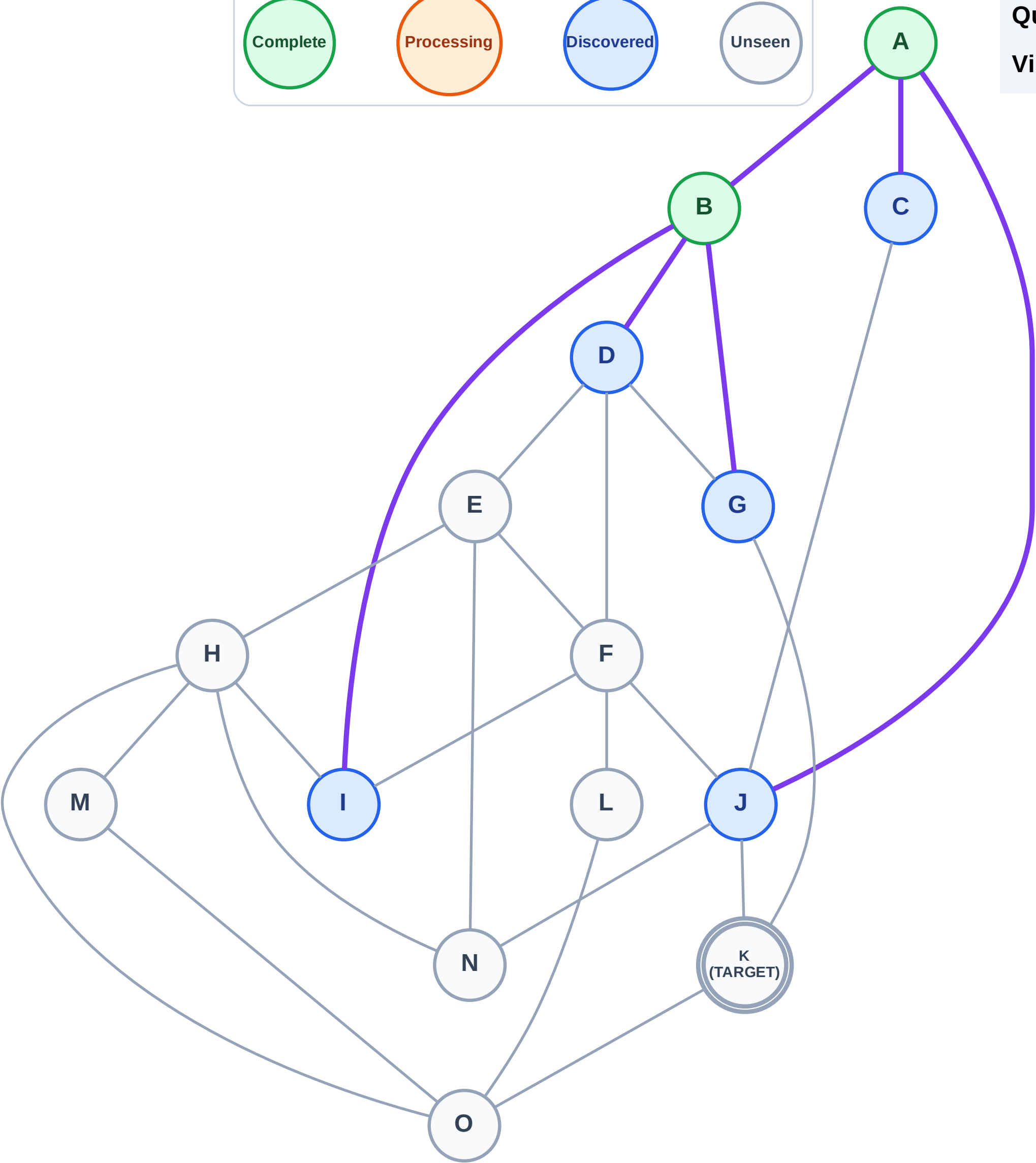
Step 5: Discover D, G, I from B

Legend



Queue: C → J → D → G → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

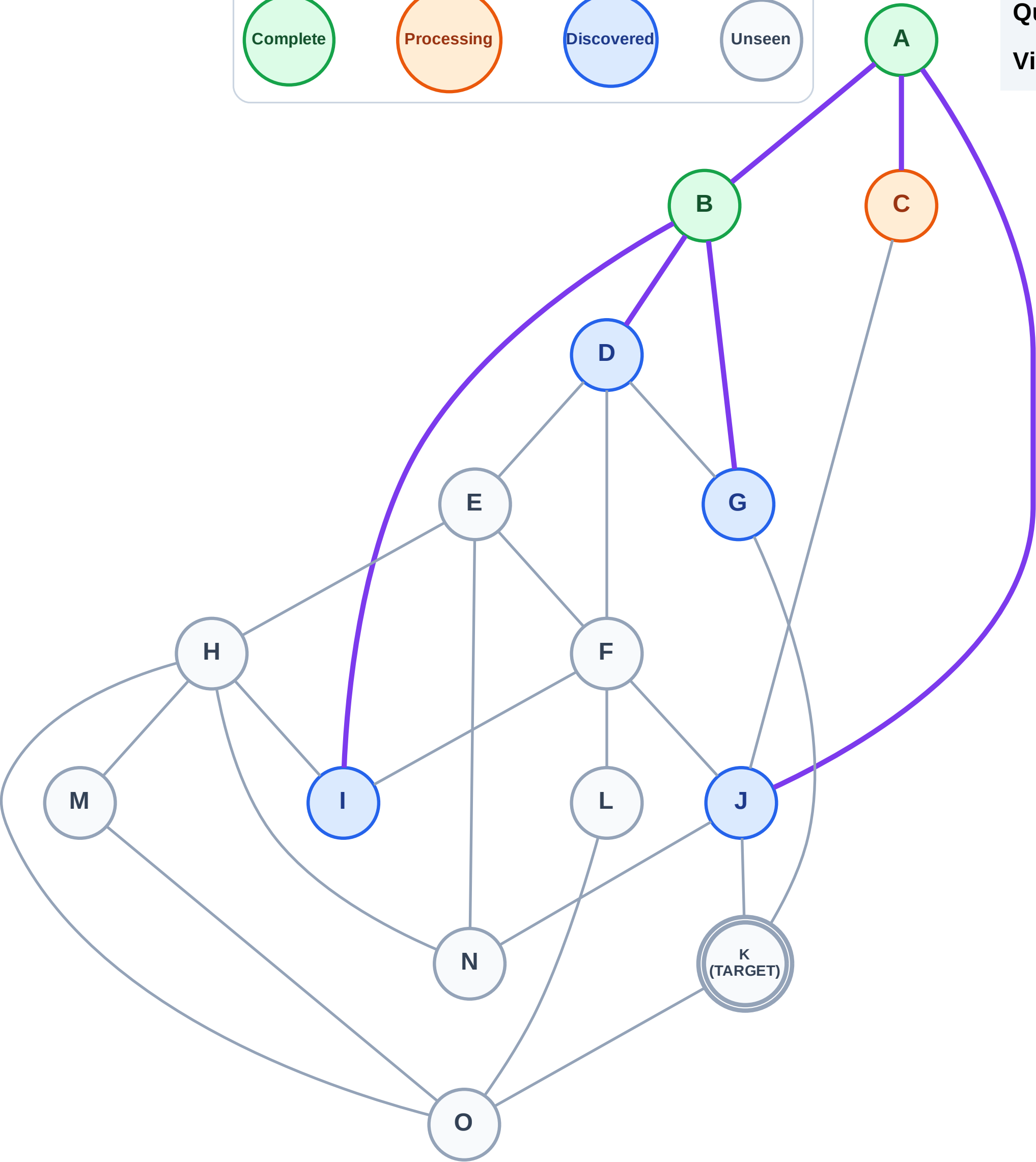
Processing

Discovered

Unseen

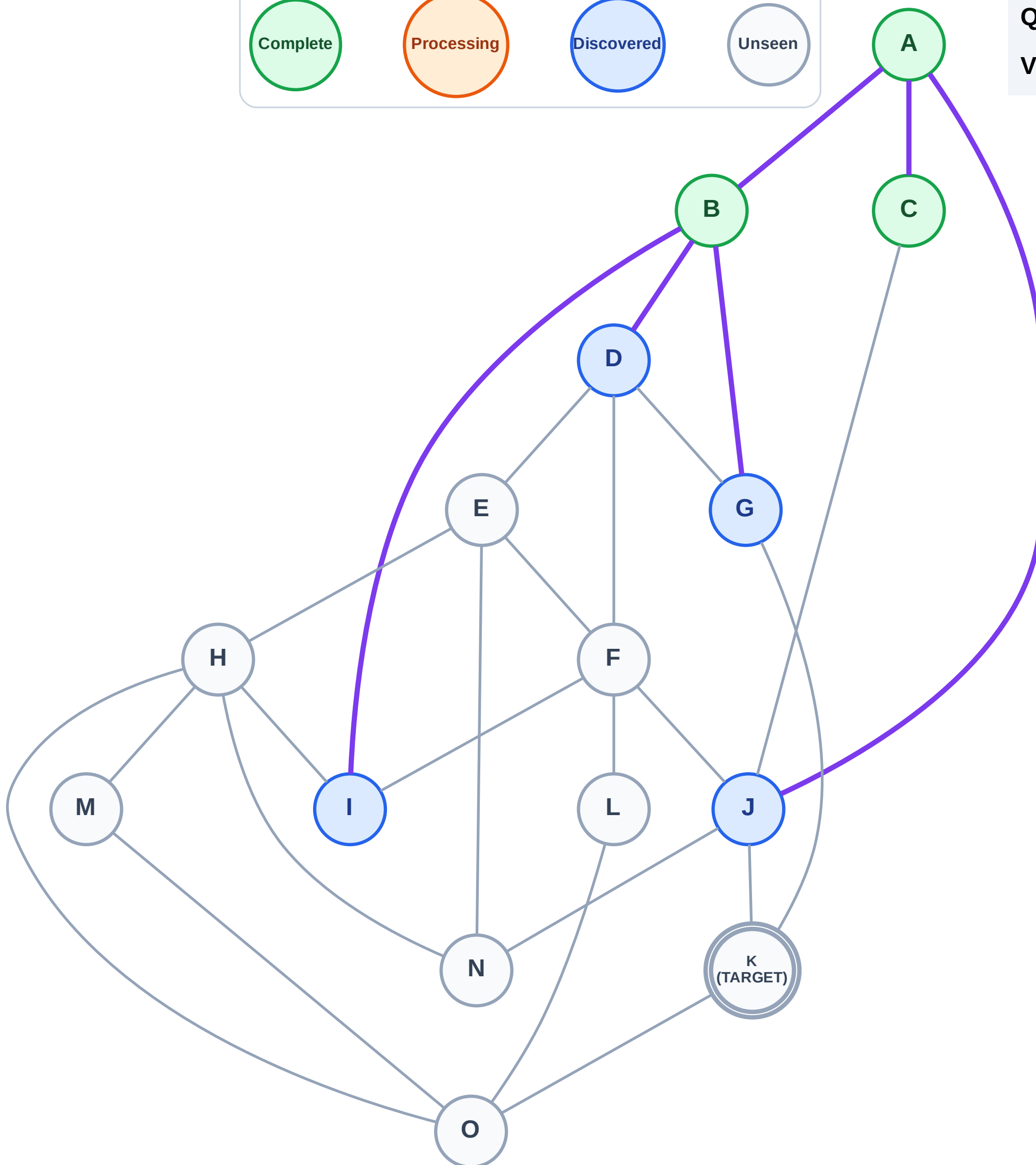
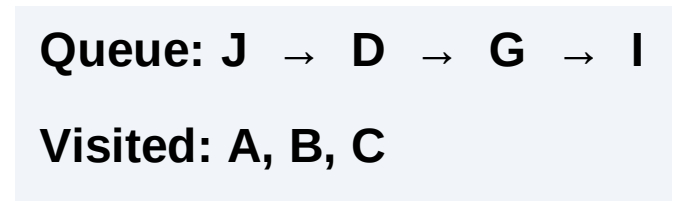
Queue: J → D → G → I

Visited: A, B



Step 7: No new neighbors from C

Step 7: No new neighbors from C



BFS Traversal

Step 8: Process node J

Legend

Complete

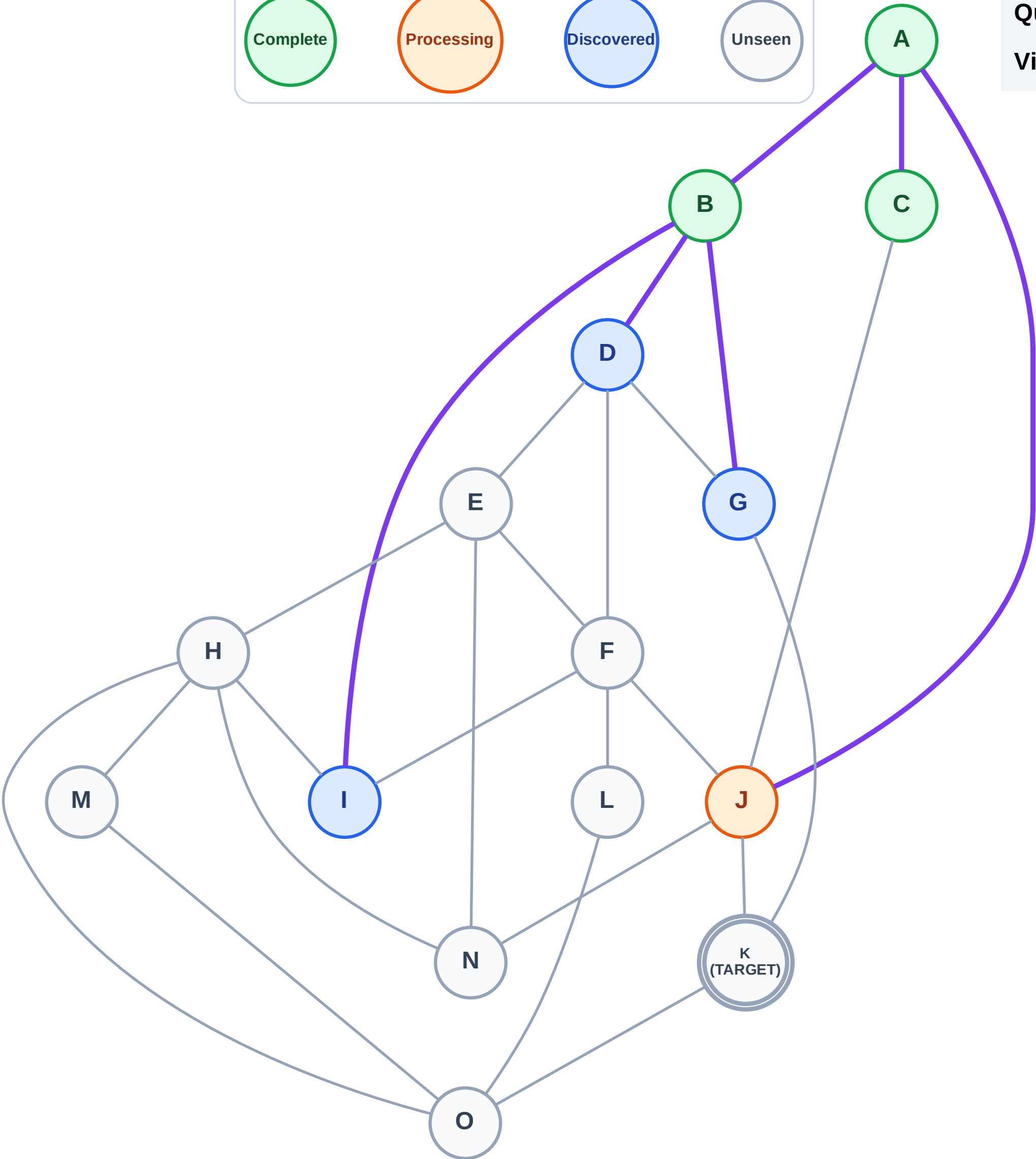
Processing

Discovered

Unseen

Queue: D → G → I

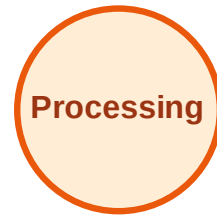
Visited: A, B, C



BFS Traversal

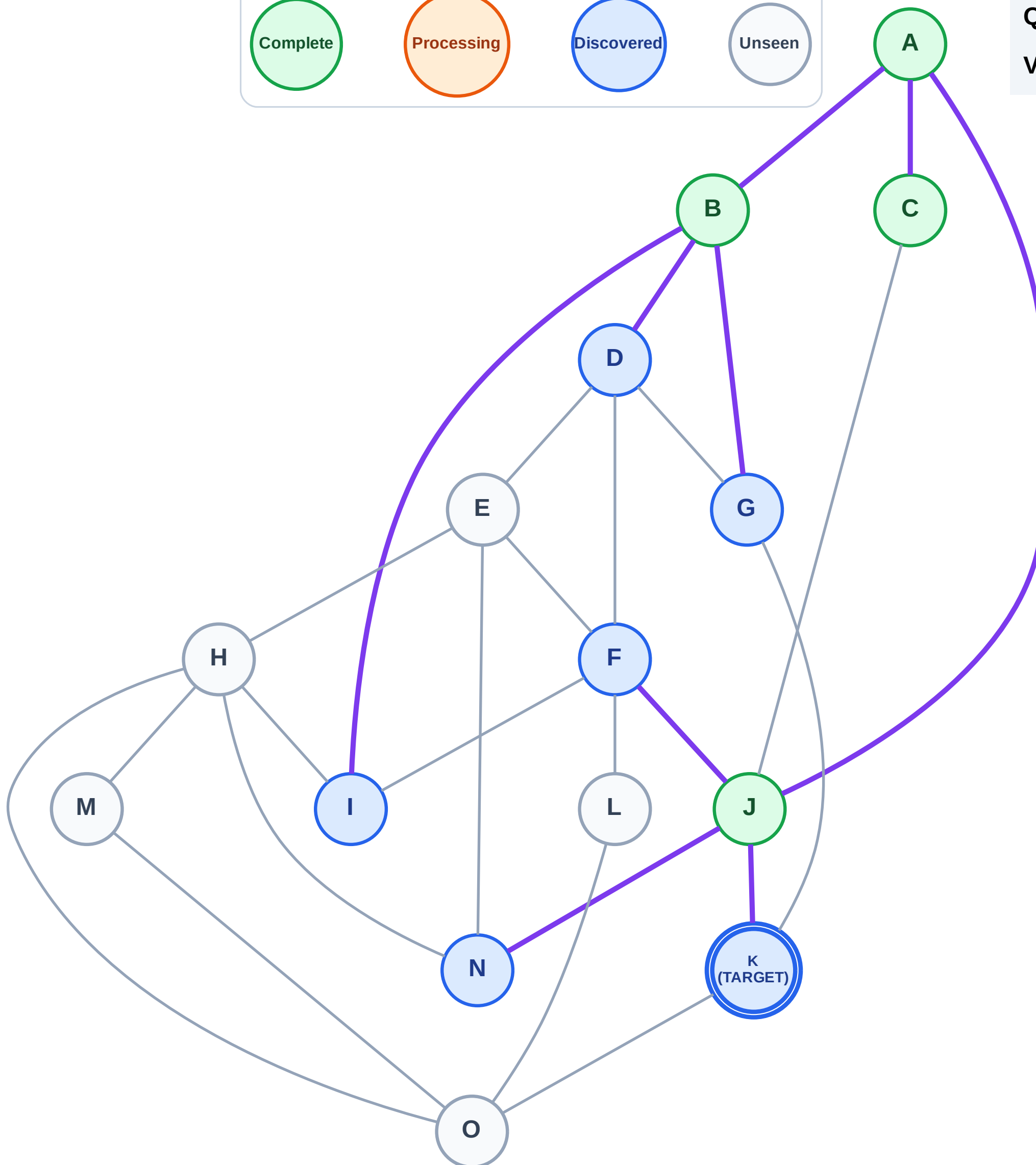
Step 9: Discover F, K, N from J

Legend



Queue: D → G → I → F → K → N

Visited: A, B, C, J



BFS Traversal

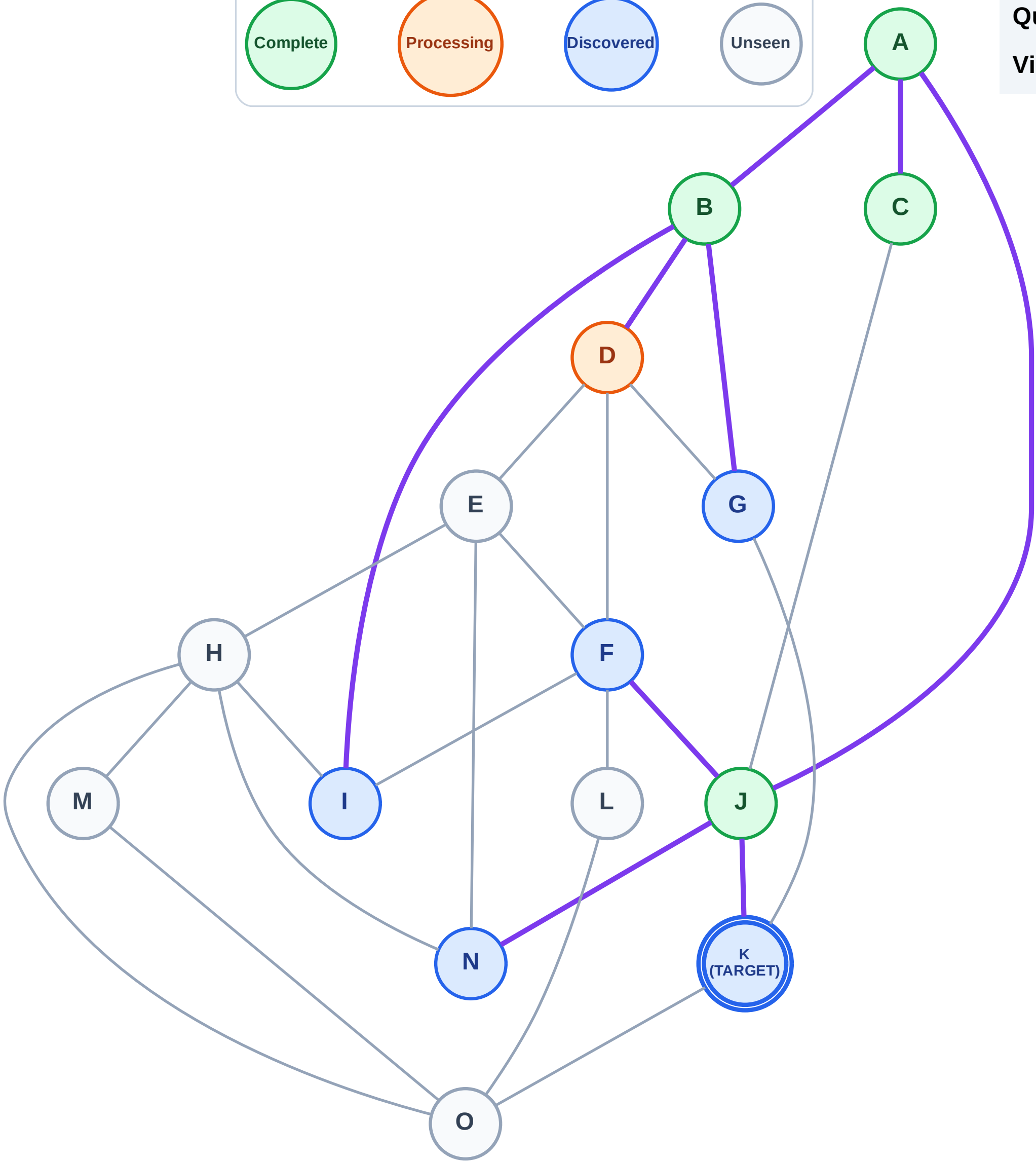
Step 10: Process node D

Legend



Queue: G → I → F → K → N

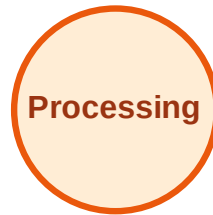
Visited: A, B, C, J



BFS Traversal

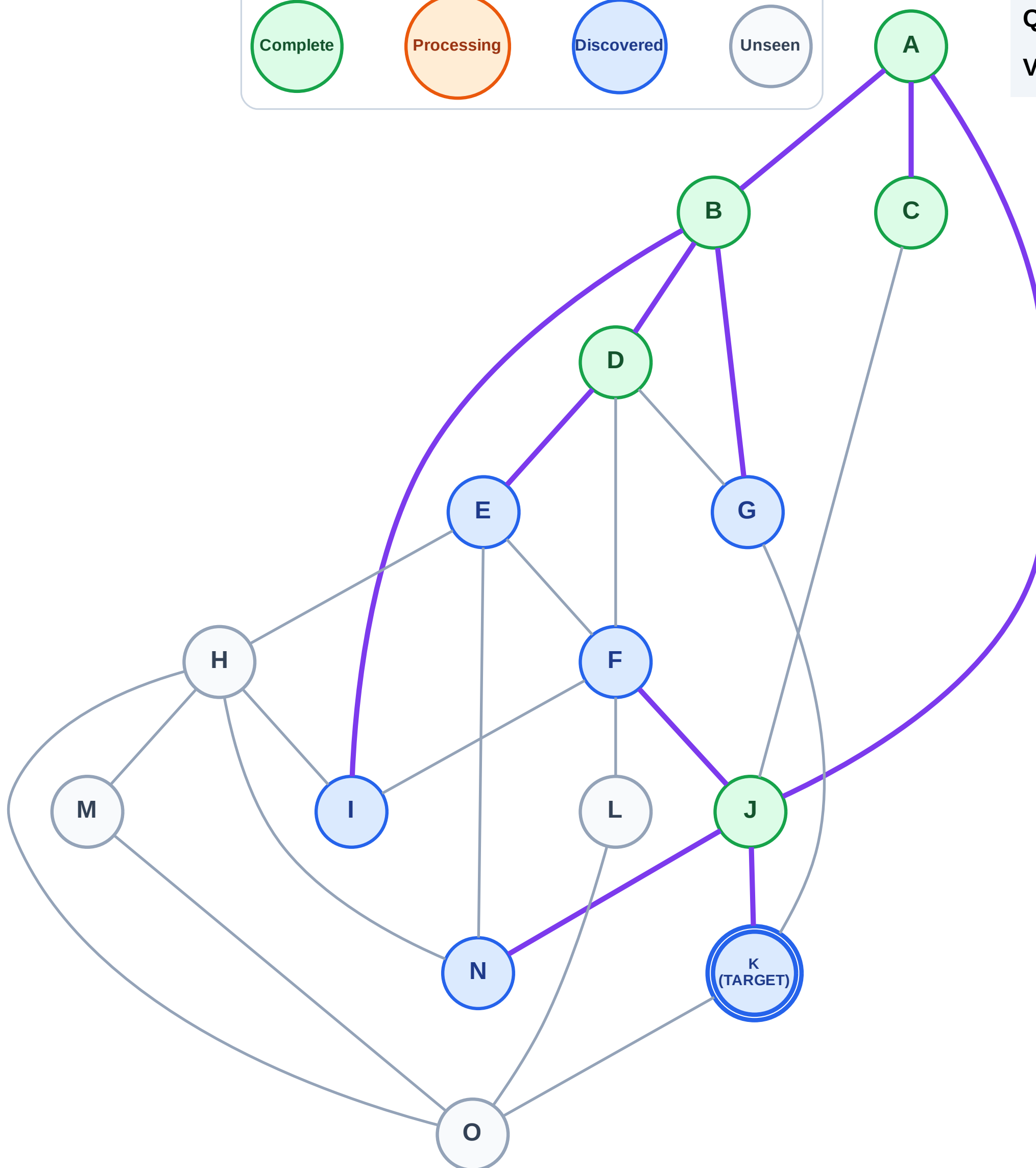
Step 11: Discover E from D

Legend



Queue: G → I → F → K → N → E

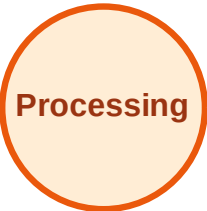
Visited: A, B, C, D, J



BFS Traversal

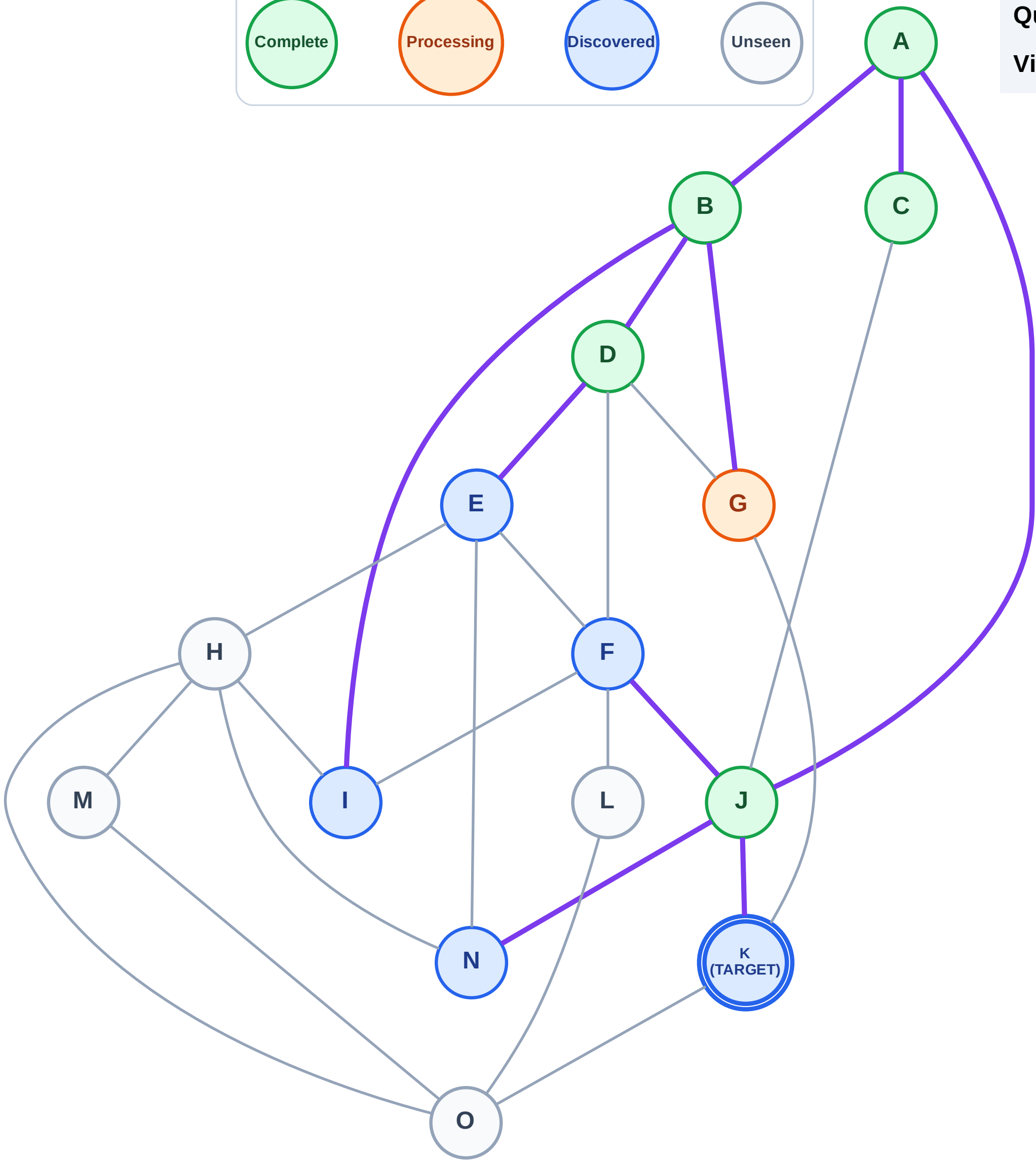
Step 12: Process node G

Legend



Queue: I → F → K → N → E

Visited: A, B, C, D, J



BFS Traversal

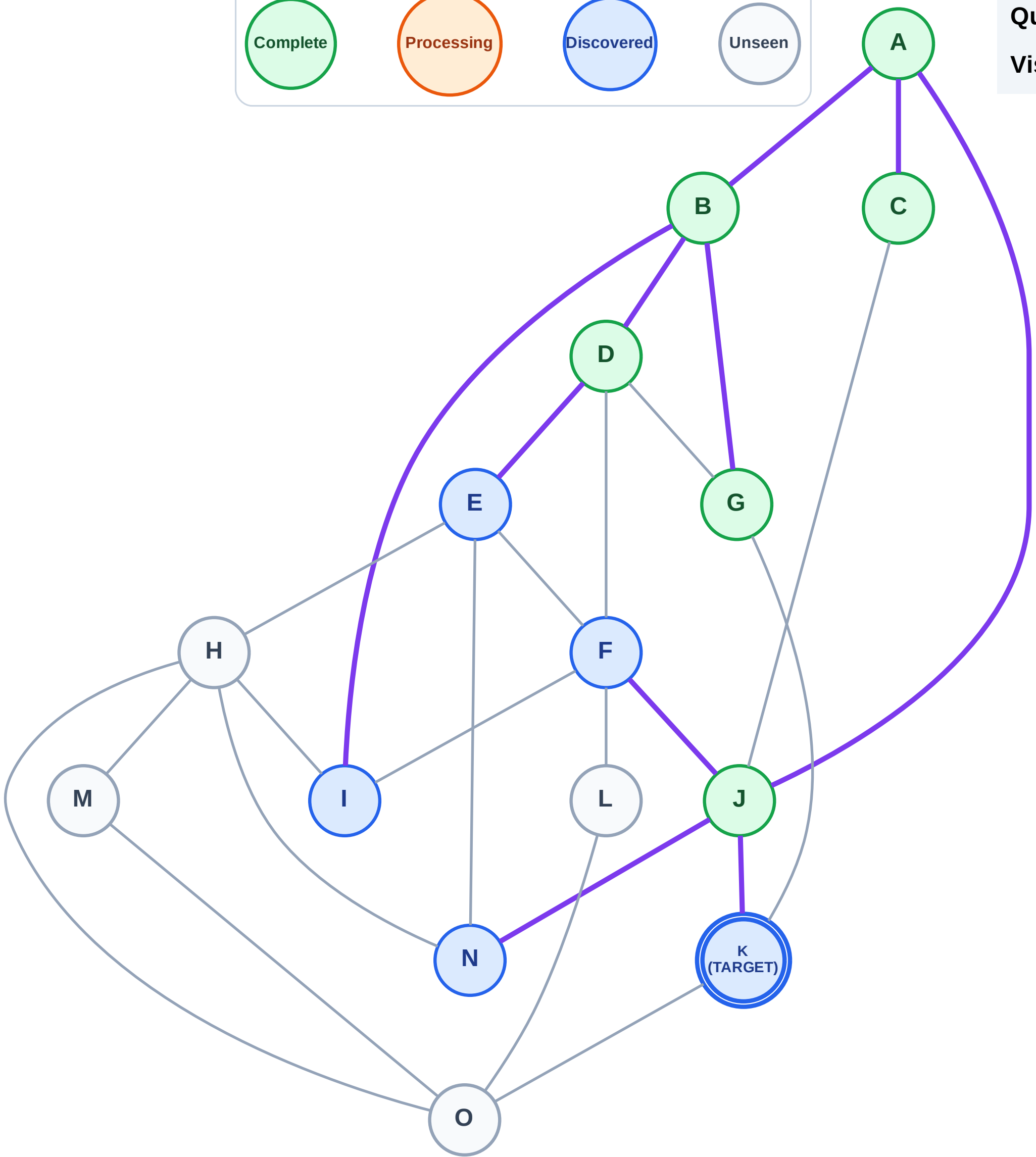
Step 13: No new neighbors from G

Legend



Queue: I → F → K → N → E

Visited: A, B, C, D, G, J



BFS Traversal

Step 14: Process node I

Legend

Complete

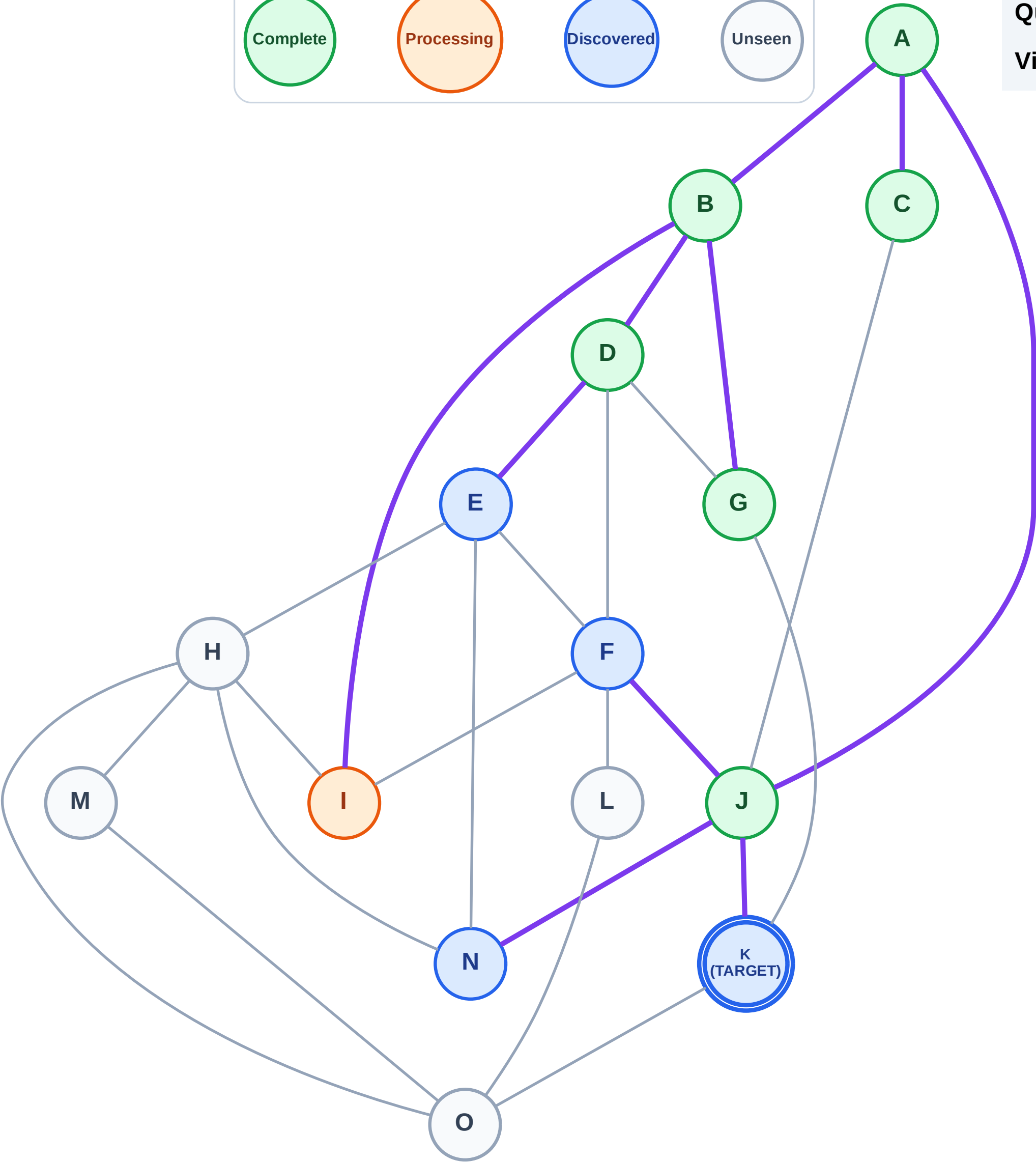
Processing

Discovered

Unseen

Queue: F → K → N → E

Visited: A, B, C, D, G, J



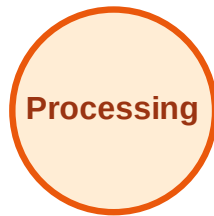
BFS Traversal

Step 15: Discover H from I

Legend



Complete



Processing



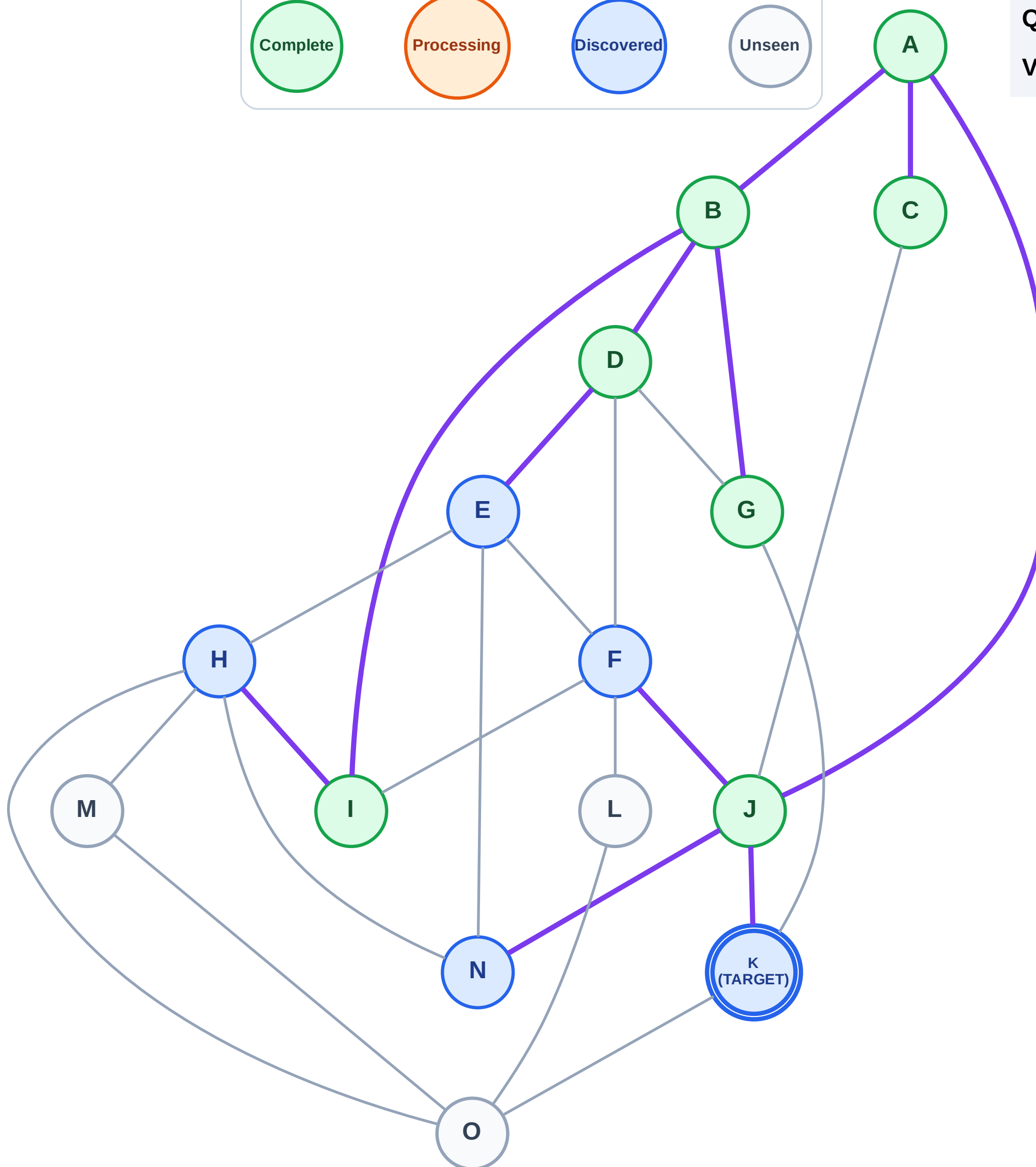
Discovered



Unseen

Queue: F → K → N → E → H

Visited: A, B, C, D, G, I, J



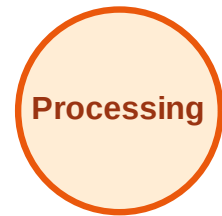
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



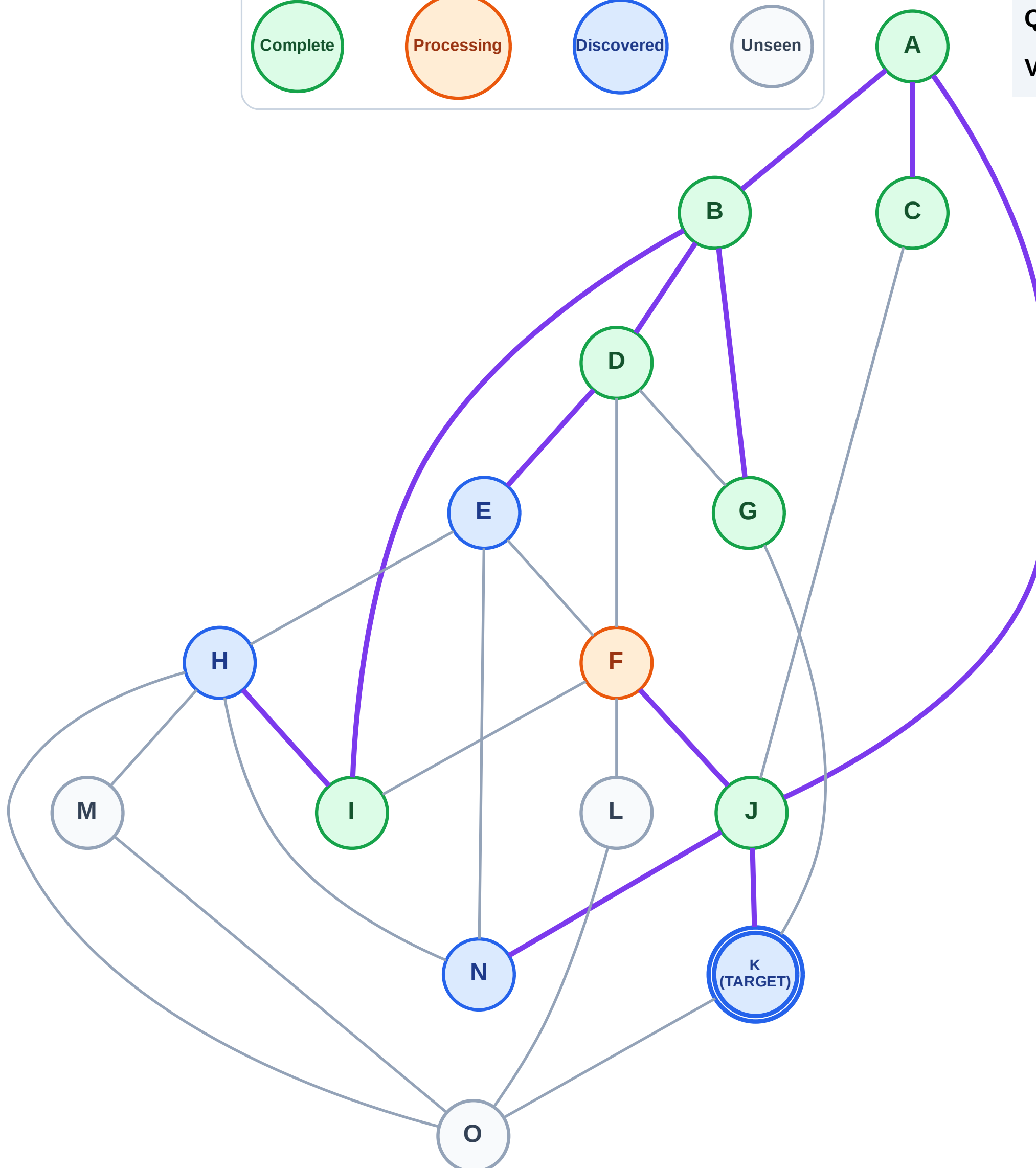
Discovered



Unseen

Queue: K → N → E → H

Visited: A, B, C, D, G, I, J



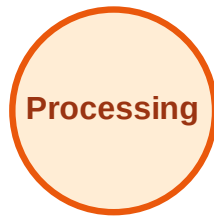
BFS Traversal

Step 17: Discover L from F

Legend



Complete



Processing



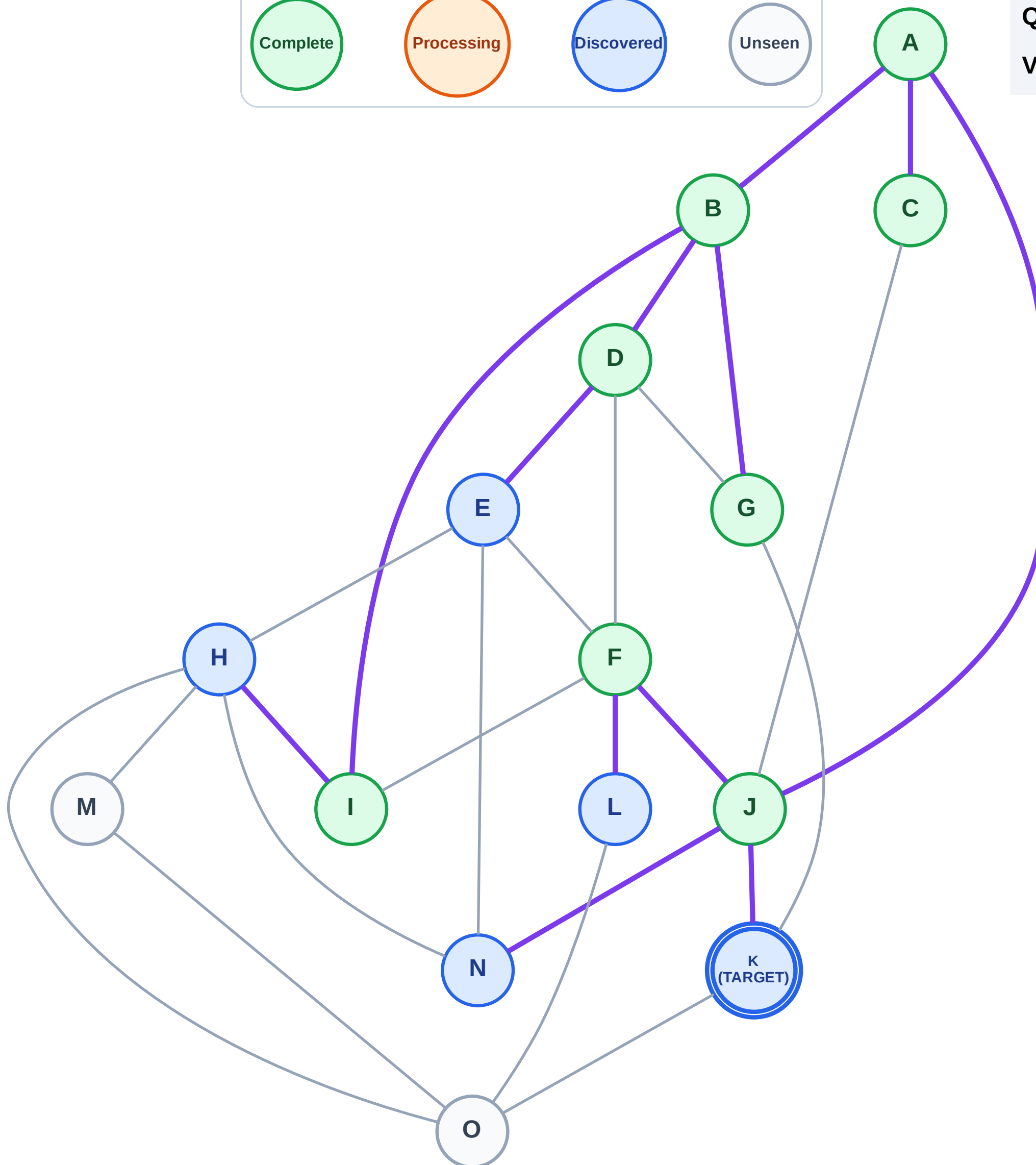
Discovered



Unseen

Queue: K → N → E → H → L

Visited: A, B, C, D, F, G, I, J



BFS Traversal

Step 18: Process node K

Legend



Complete



Processing



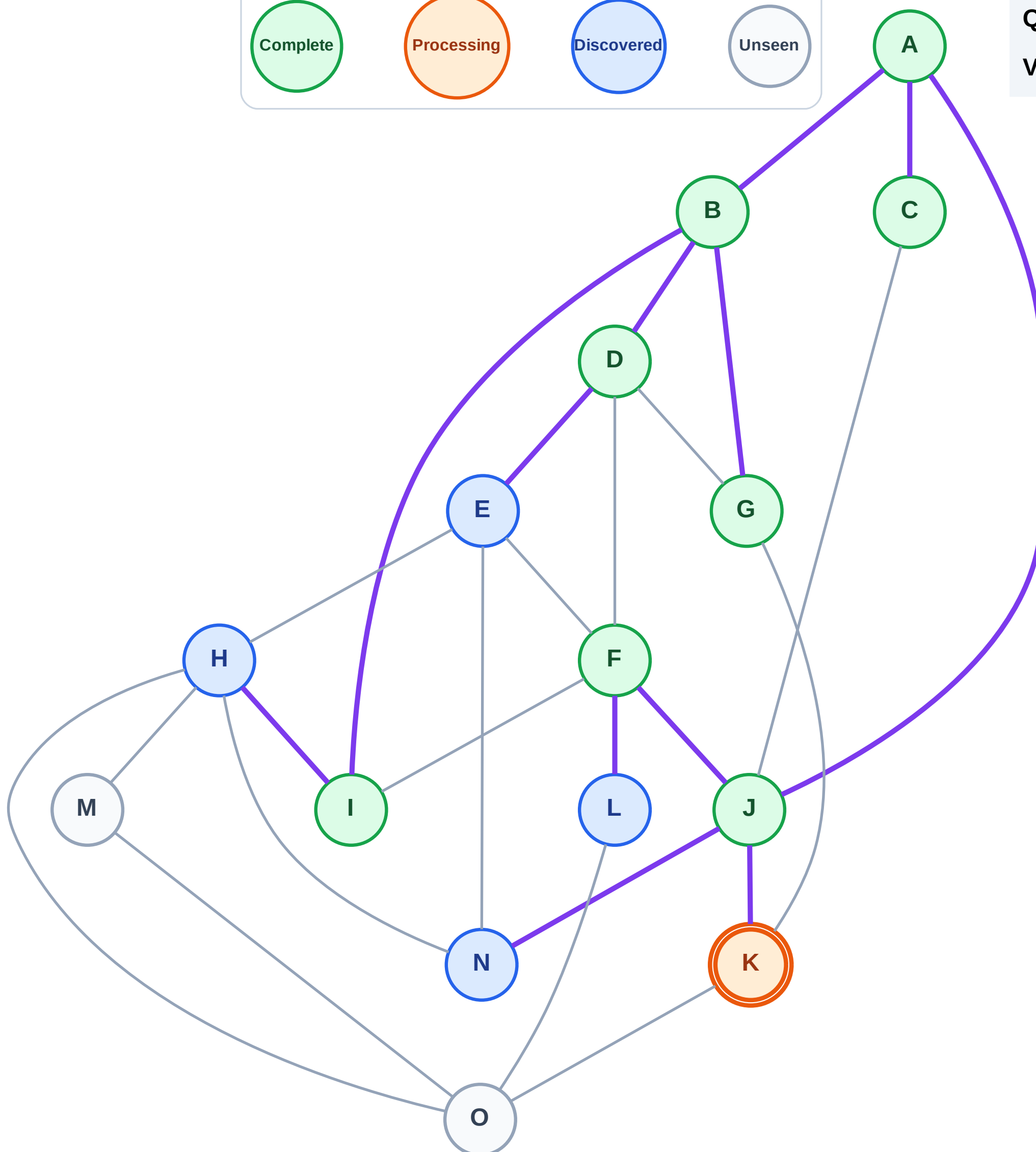
Discovered



Unseen

Queue: N → E → H → L

Visited: A, B, C, D, F, G, I, J



BFS Traversal

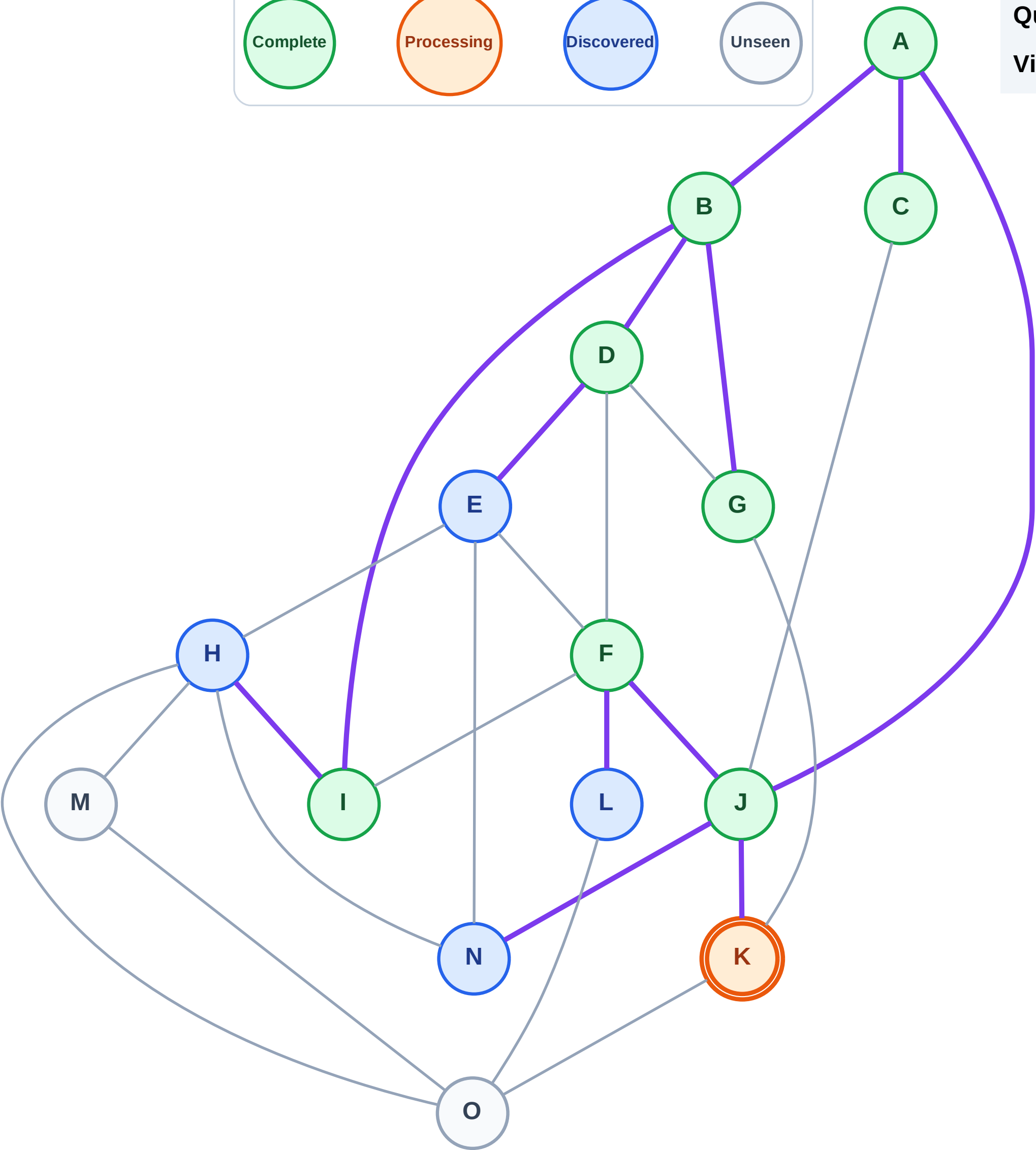
Step 19: TARGET NODE K REACHED! Path (2 hops): A → J → K

Legend



Queue: N → E → H → L

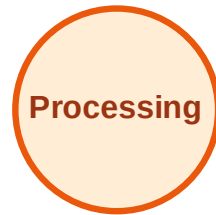
Visited: A, B, C, D, F, G, I, J, K



DFS Traversal

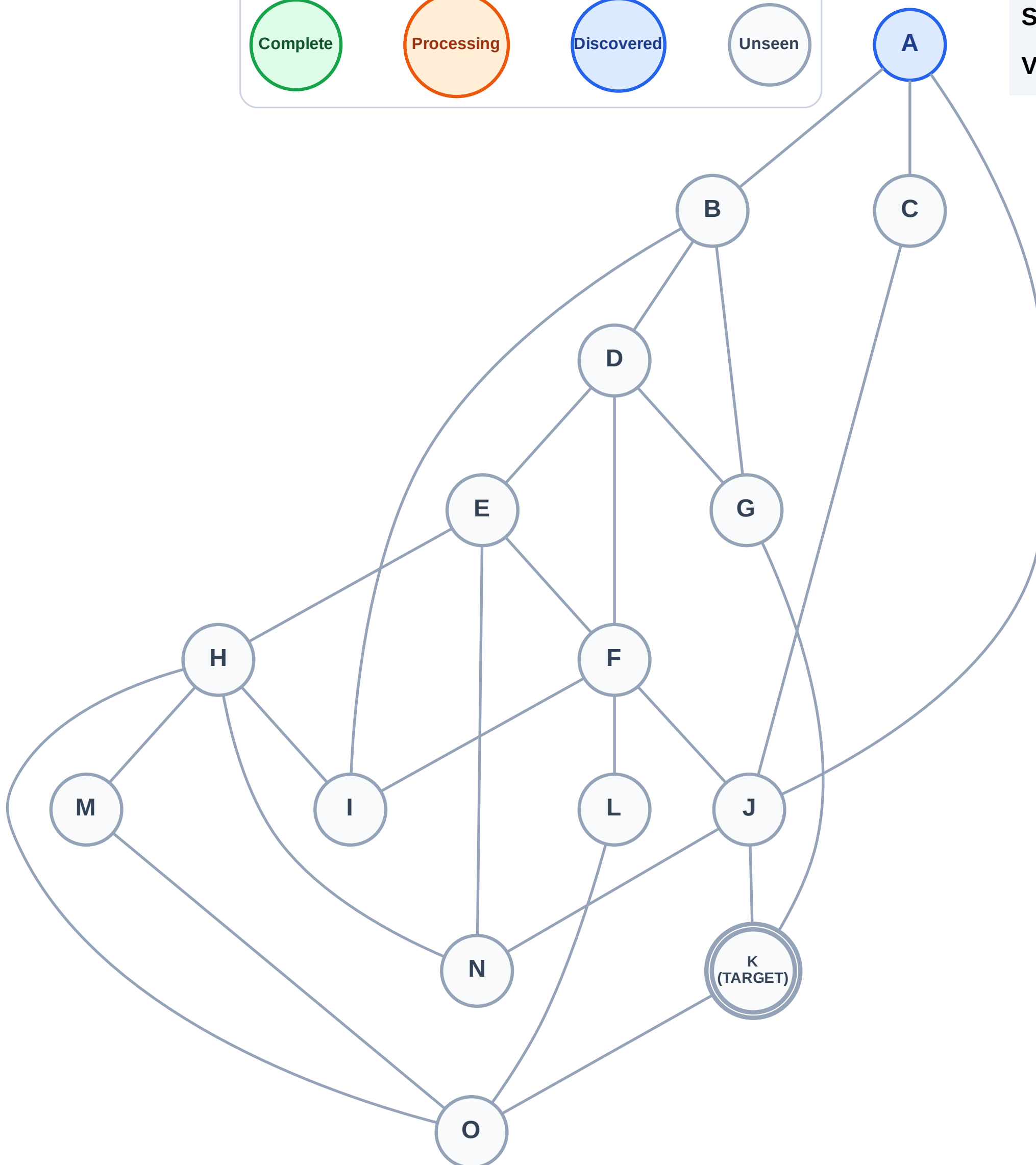
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

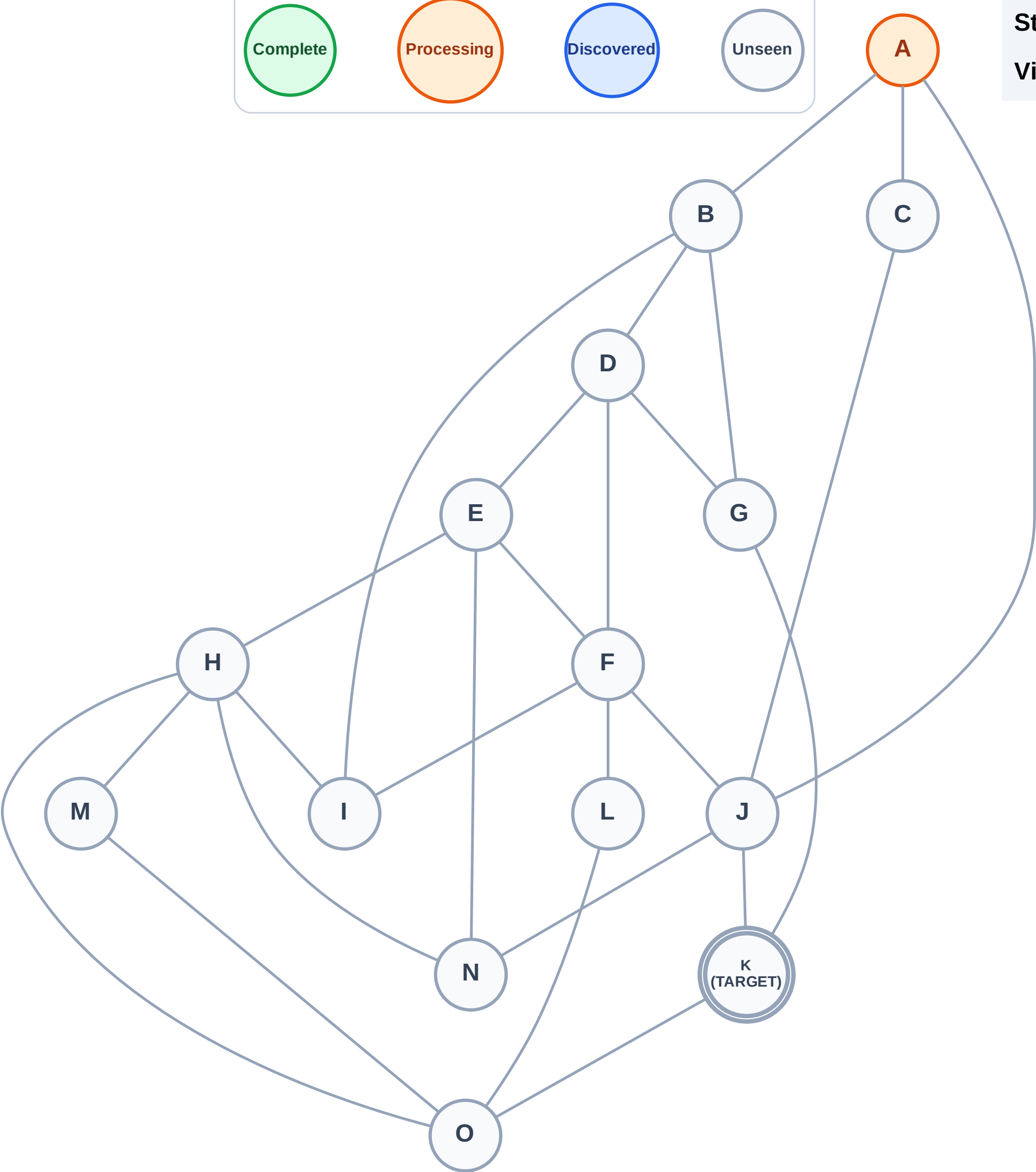
Complete

Processing

Discovered

Unseen

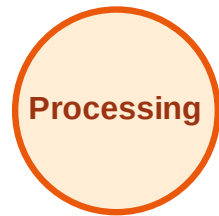
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

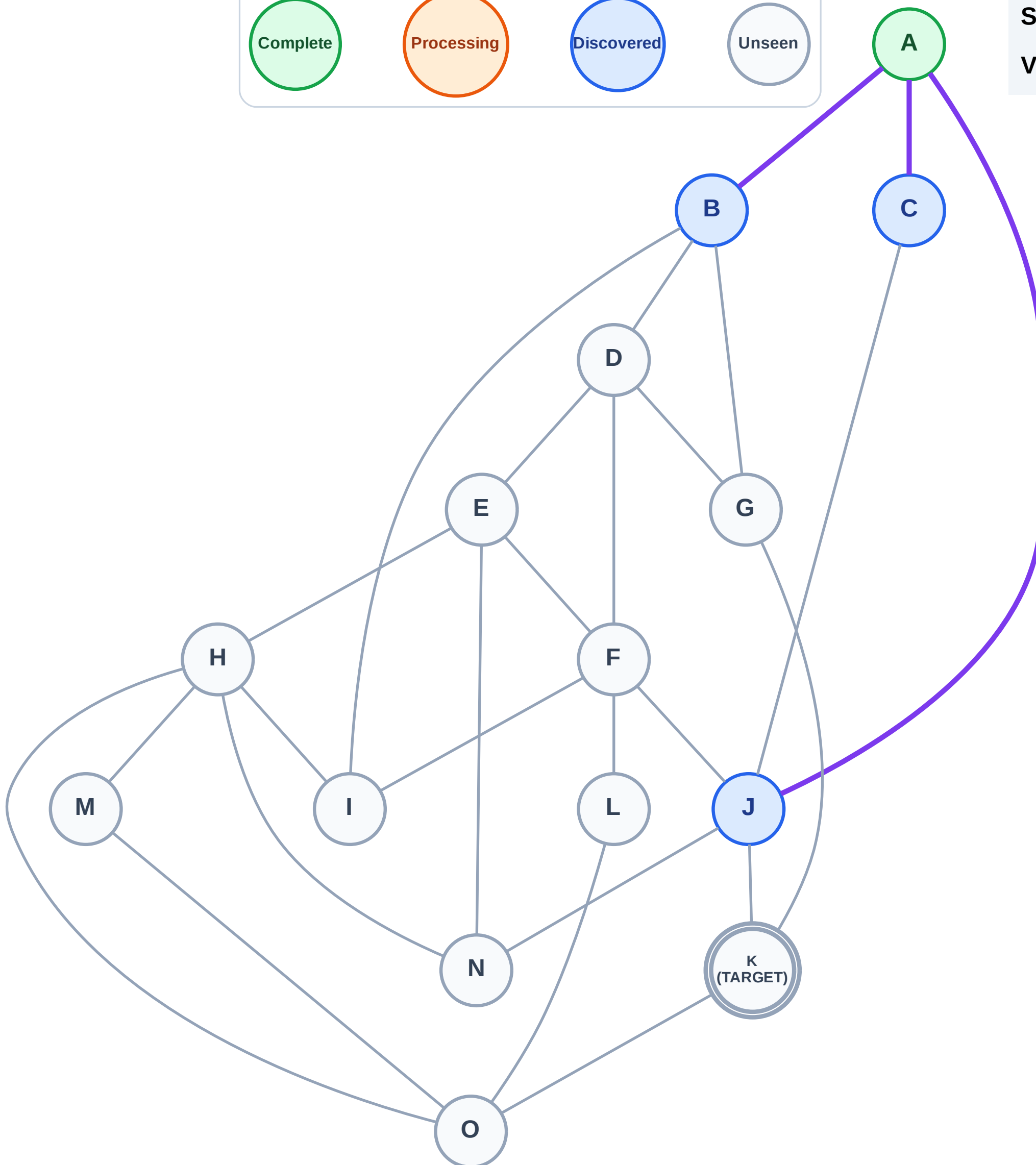
Step 3: Discover J, C, B from A

Legend



Stack (top at right): J | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

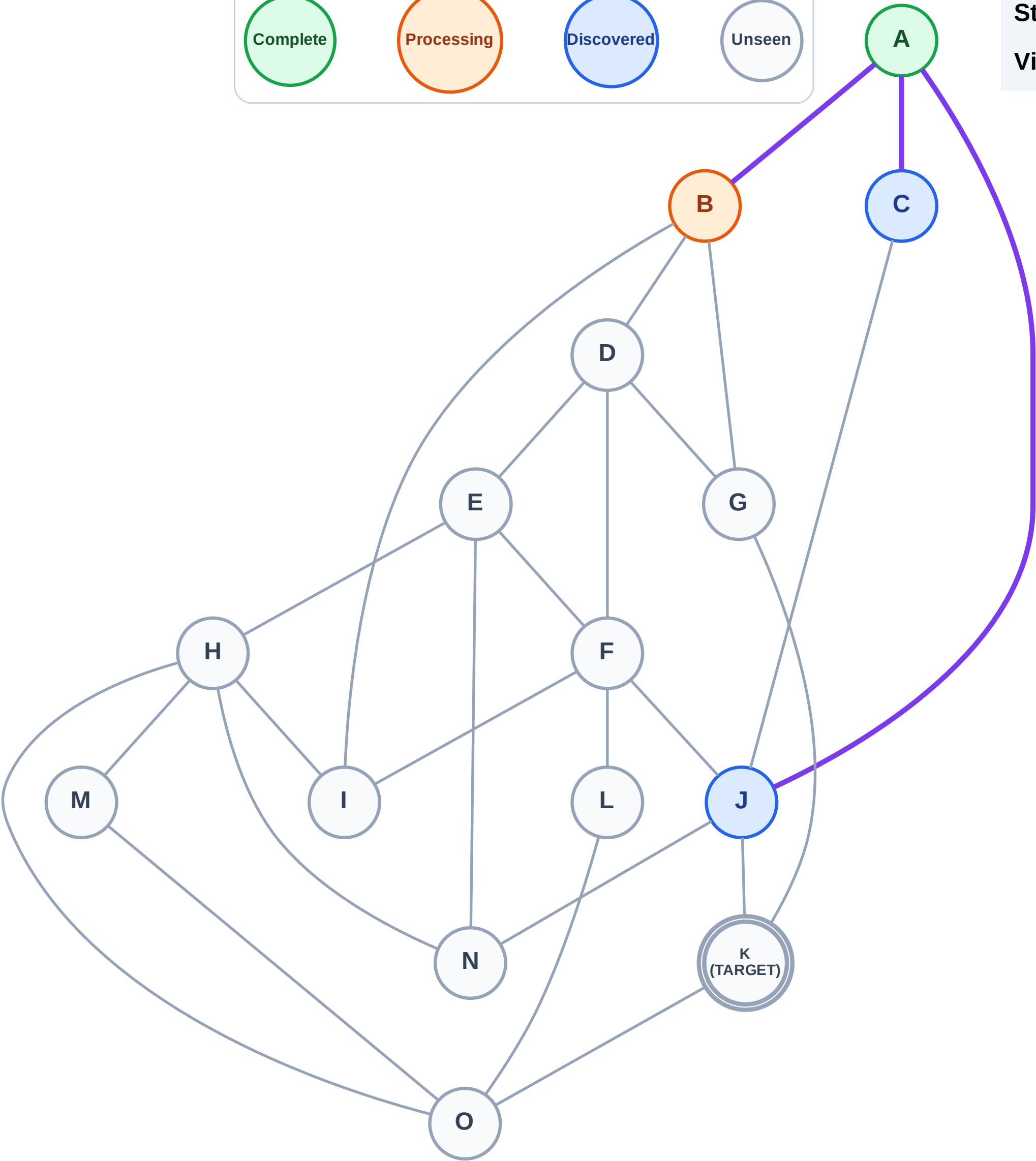
Processing

Discovered

Unseen

Stack (top at right): J | C

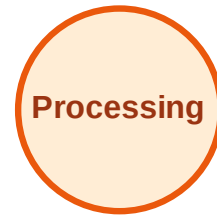
Visited: A



DFS Traversal

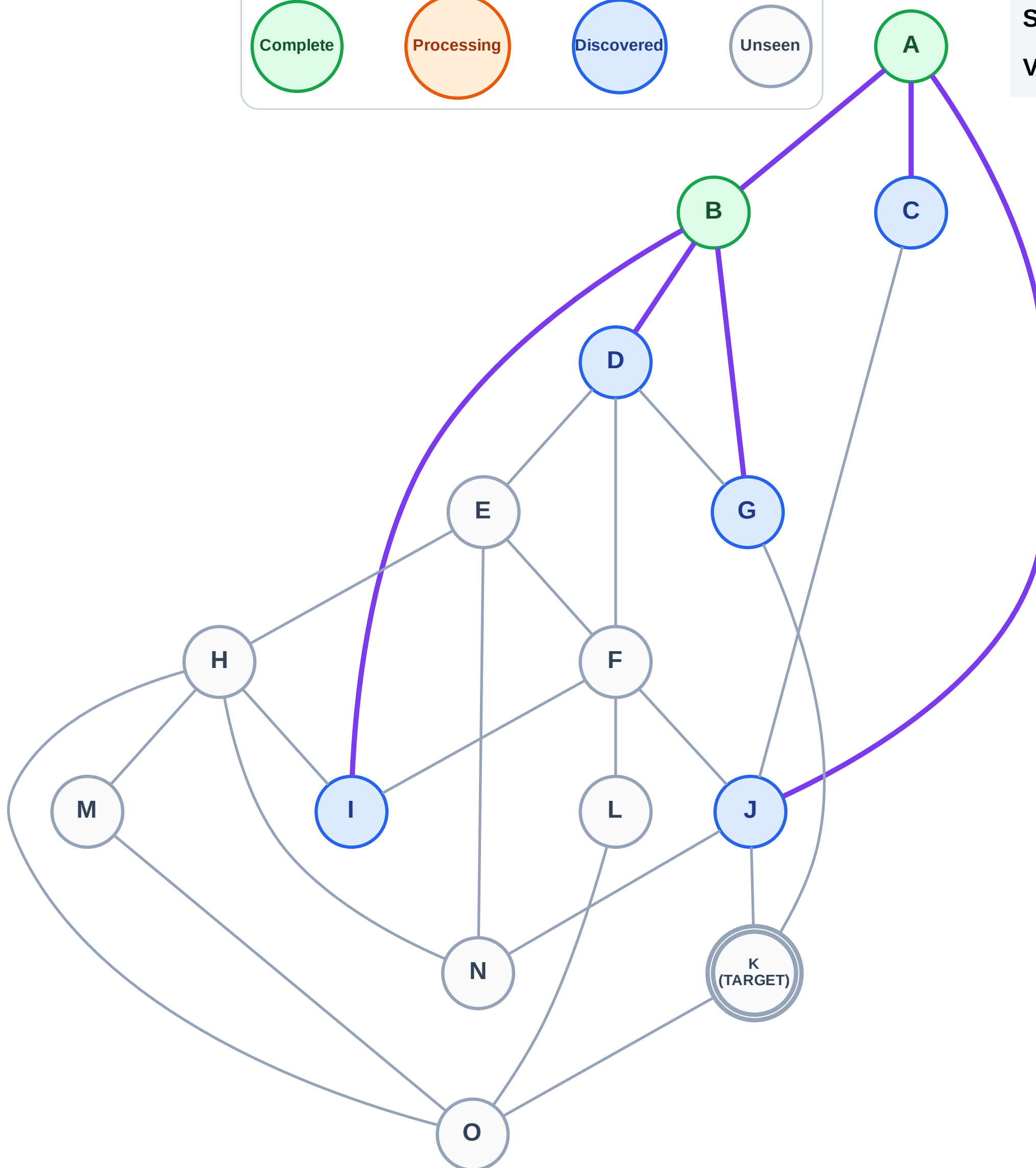
Step 5: Discover I, G, D from B

Legend



Stack (top at right): J | C | I | G | D

Visited: A, B



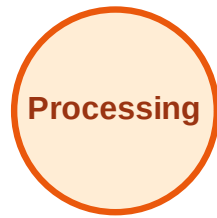
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



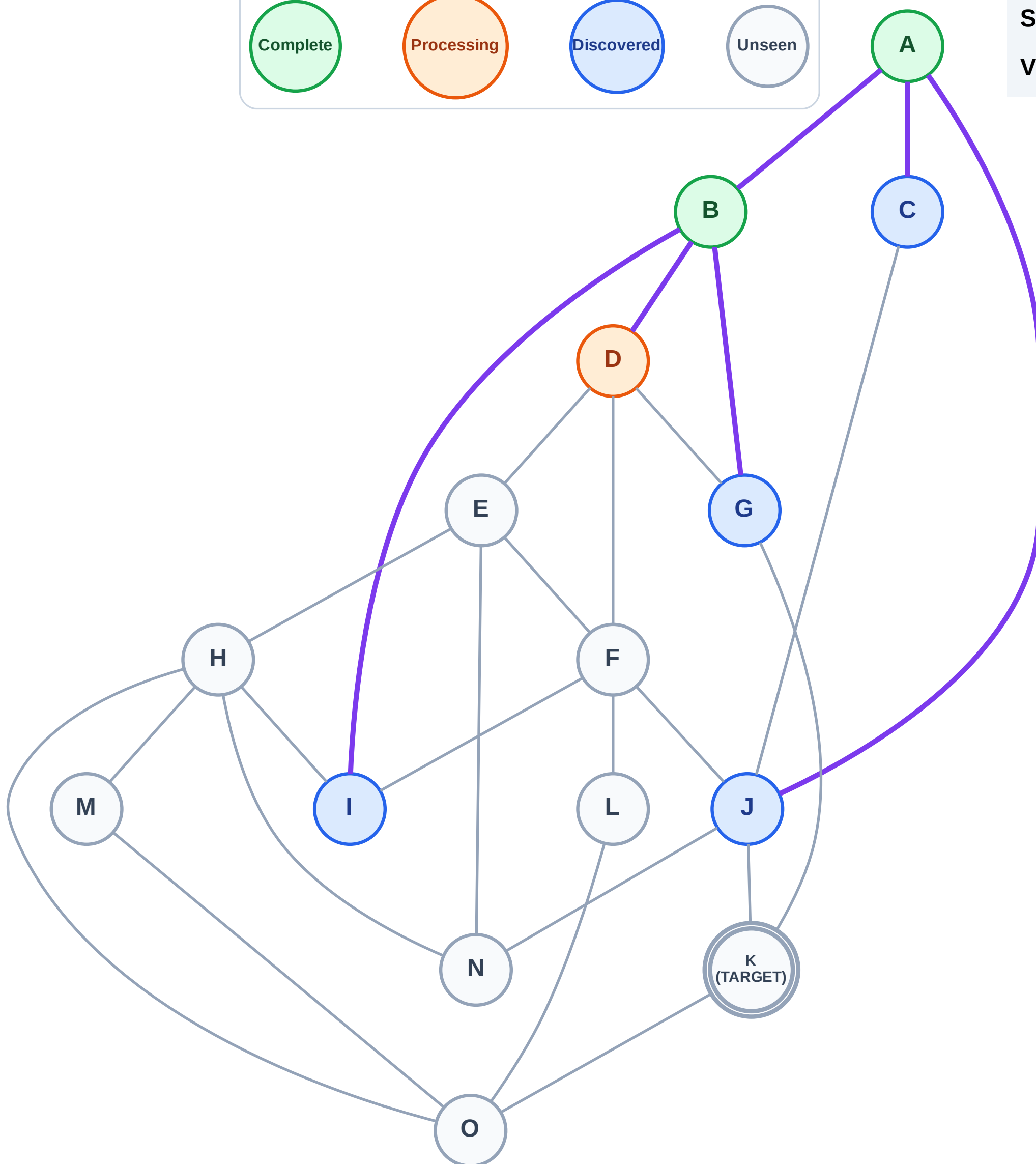
Discovered



Unseen

Stack (top at right): J | C | I | G

Visited: A, B



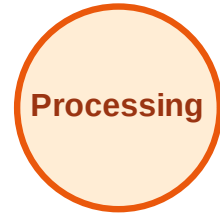
DFS Traversal

Step 7: Discover F, E from D

Legend



Complete



Processing



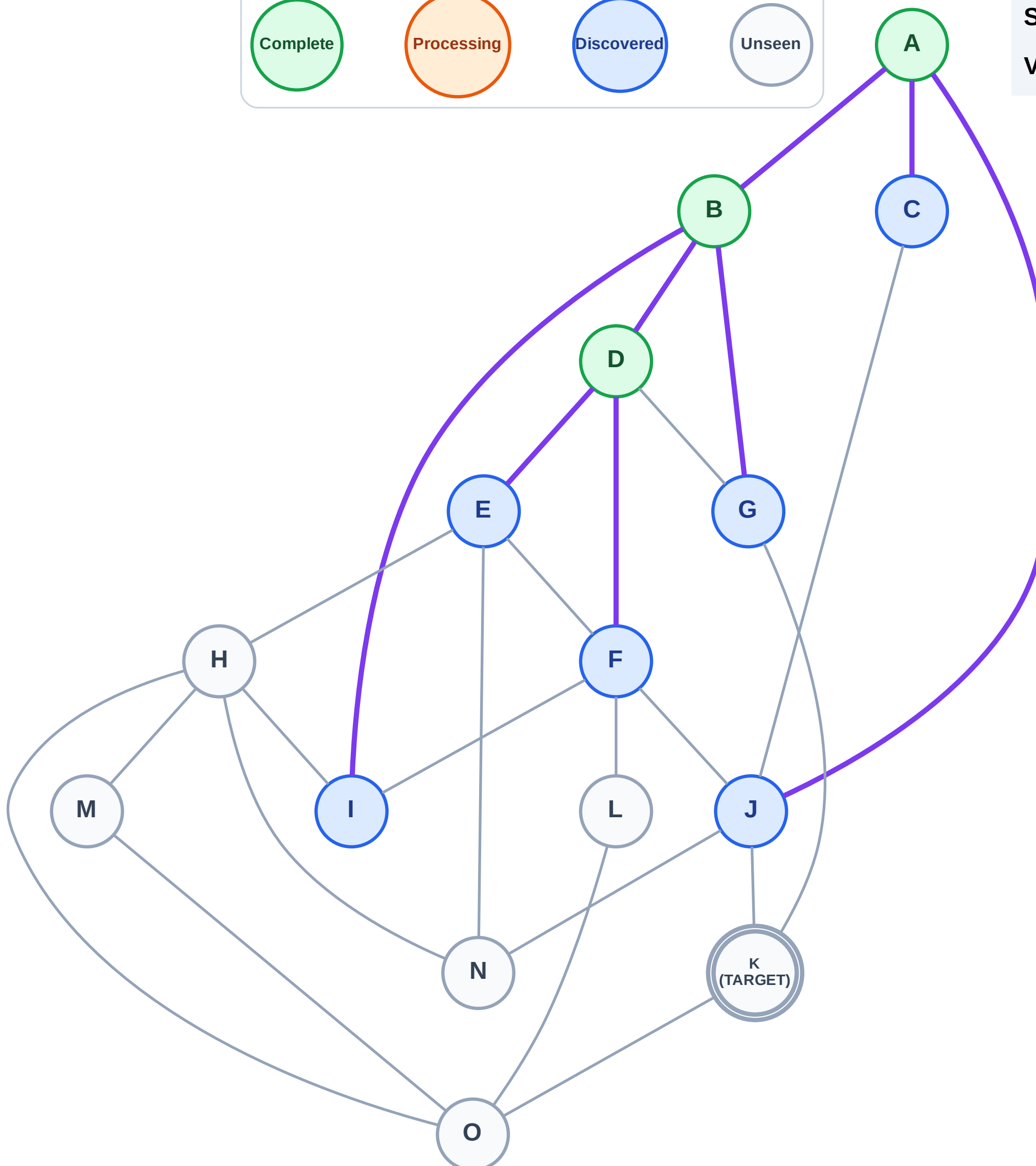
Discovered



Unseen

Stack (top at right): J | C | I | G | F | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

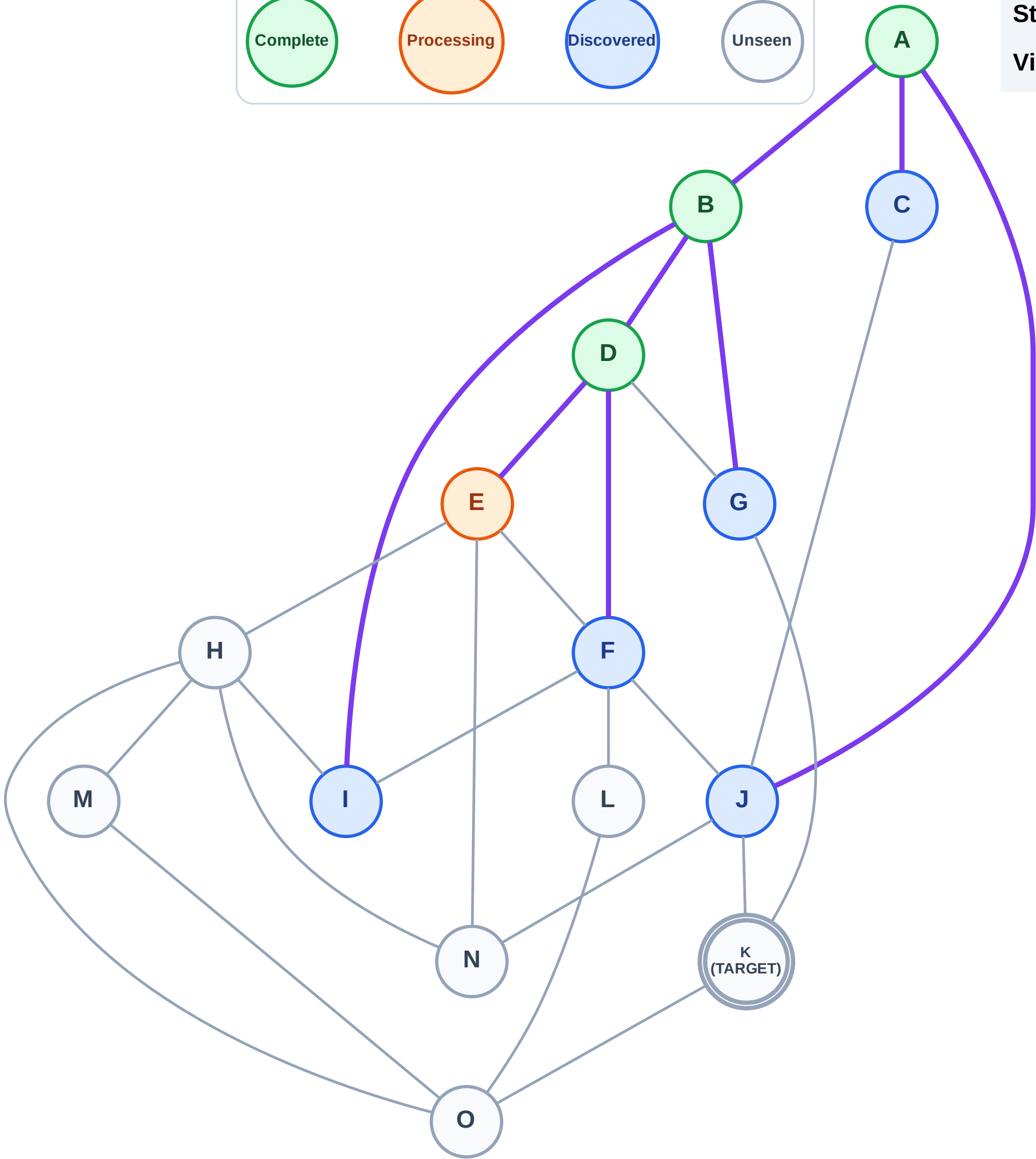
Processing

Discovered

Unseen

Stack (top at right): J | C | I | G | F

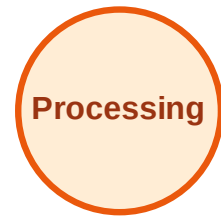
Visited: A, B, D



DFS Traversal

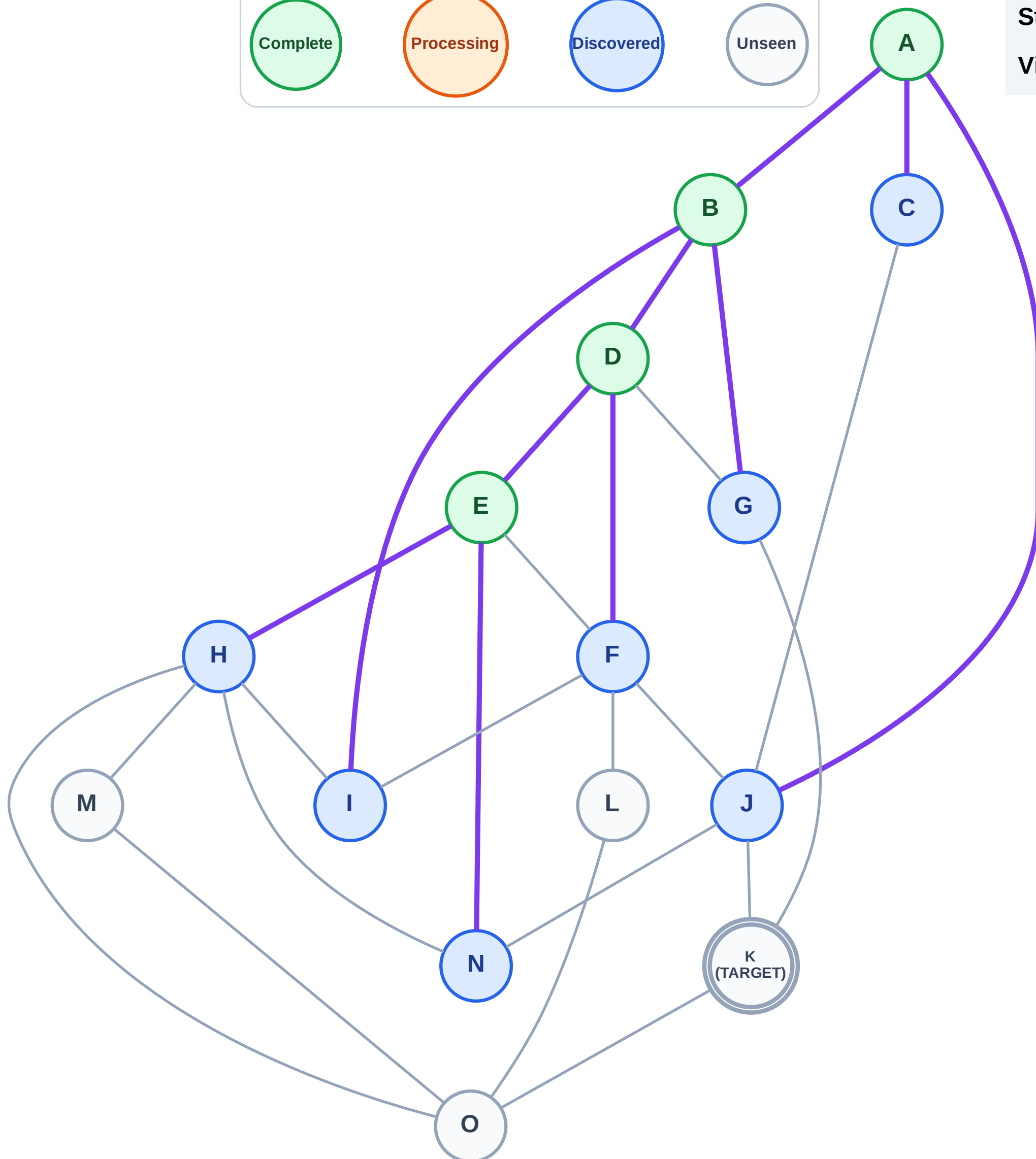
Step 9: Discover N, H from E

Legend



Stack (top at right): J | C | I | G | F | N | H

Visited: A, B, D, E



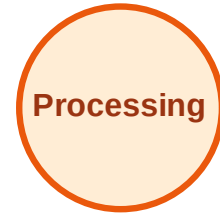
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



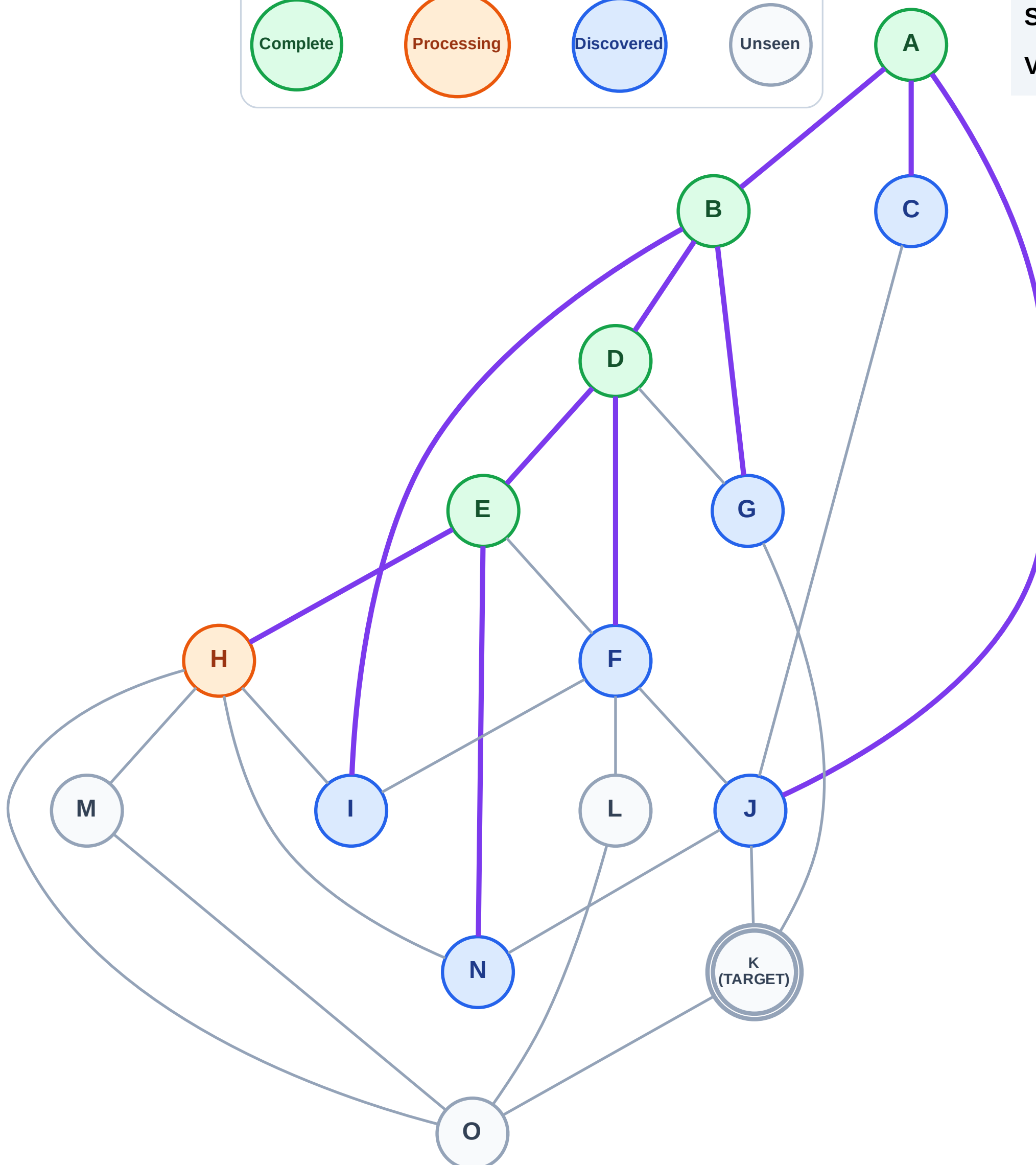
Discovered



Unseen

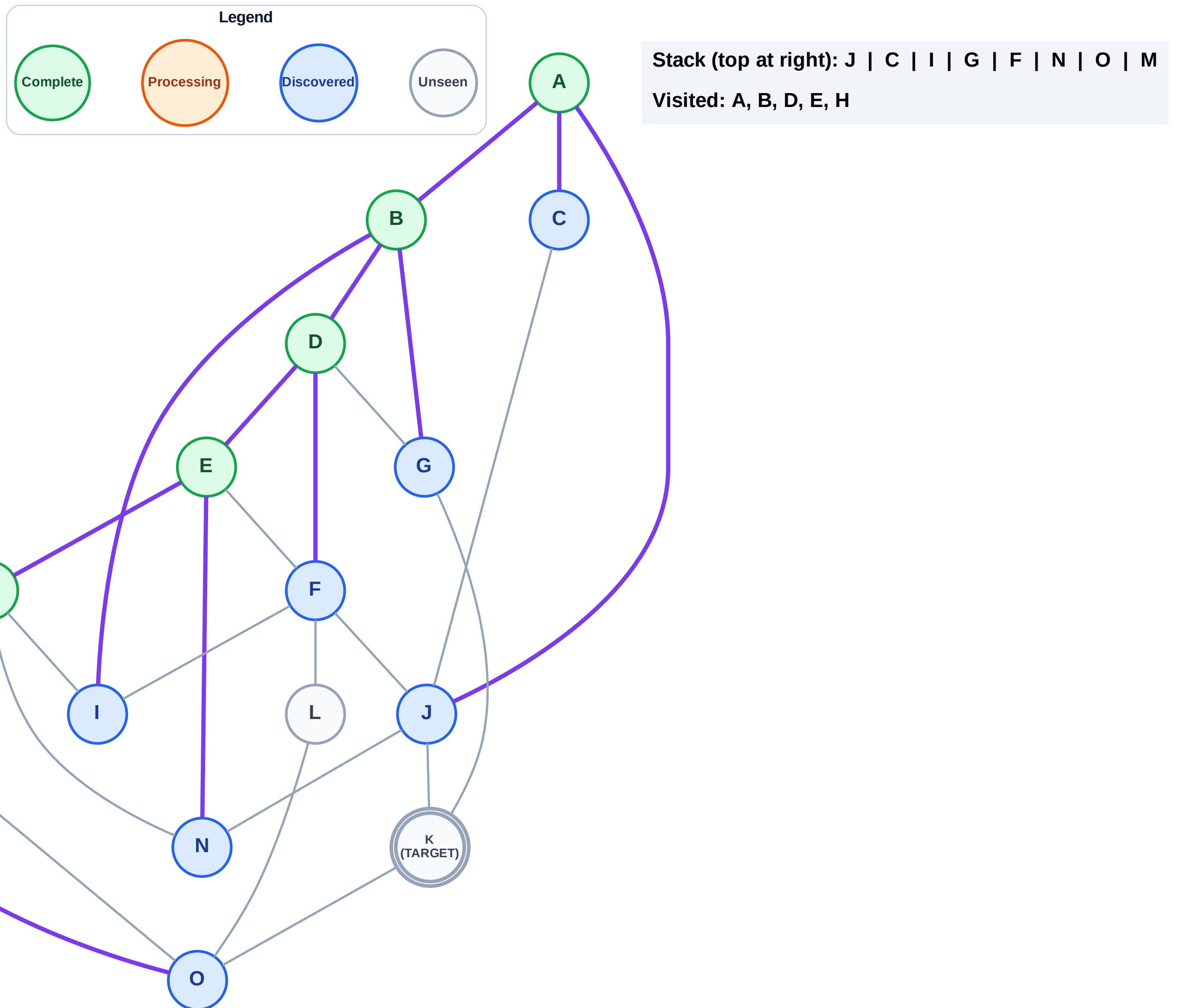
Stack (top at right): J | C | I | G | F | N

Visited: A, B, D, E



Step 11: Discover O, M from H

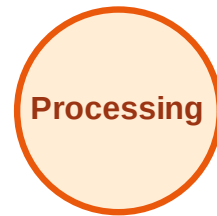
Step 11: Discover O, M from H



DFS Traversal

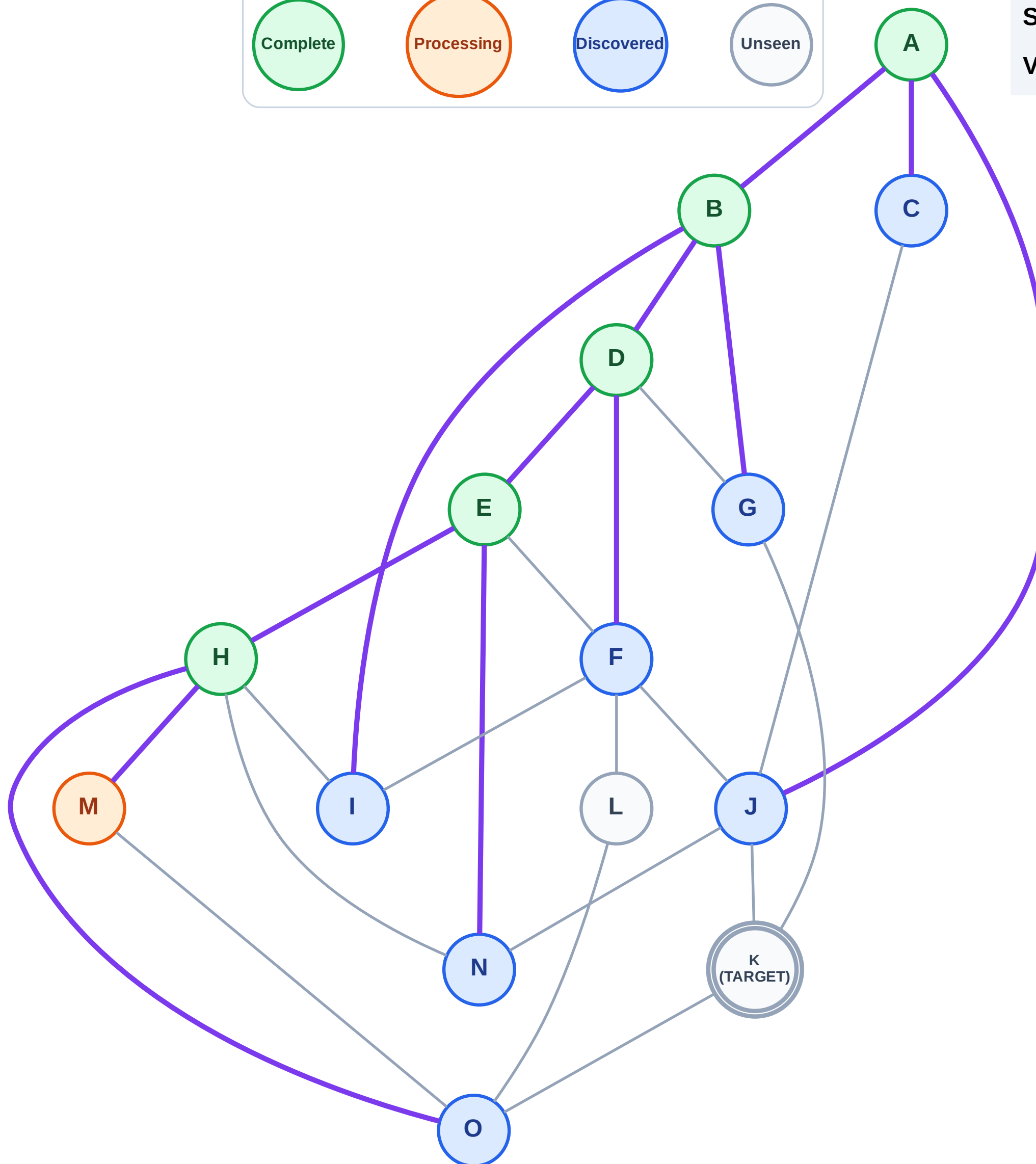
Step 12: Process node M

Legend



Stack (top at right): J | C | I | G | F | N | O

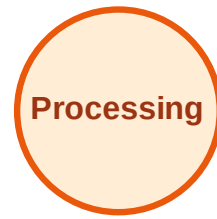
Visited: A, B, D, E, H



DFS Traversal

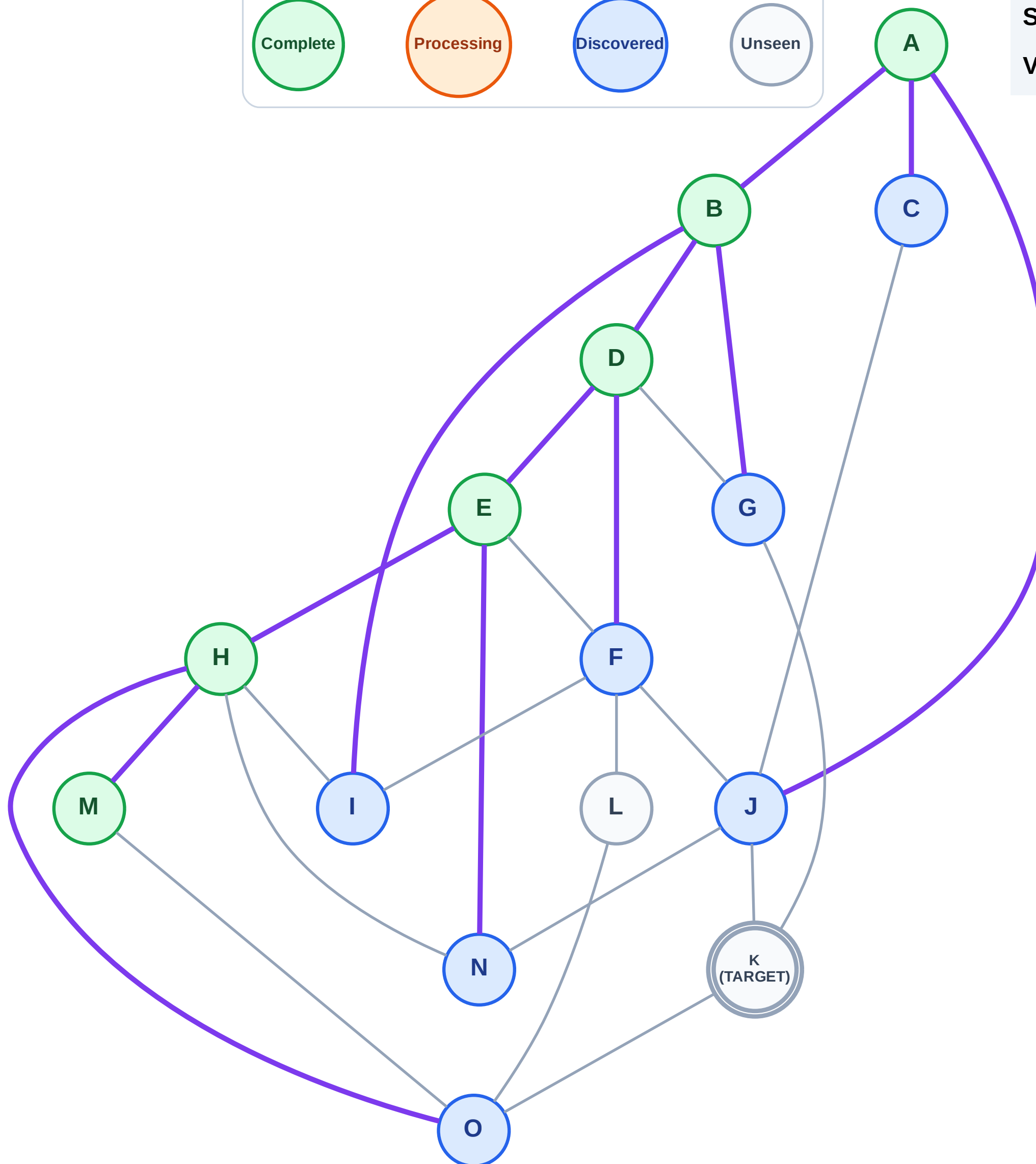
Step 13: No new neighbors from M

Legend



Stack (top at right): J | C | I | G | F | N | O

Visited: A, B, D, E, H, M



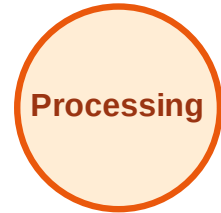
DFS Traversal

Step 14: Process node O

Legend



Complete



Processing



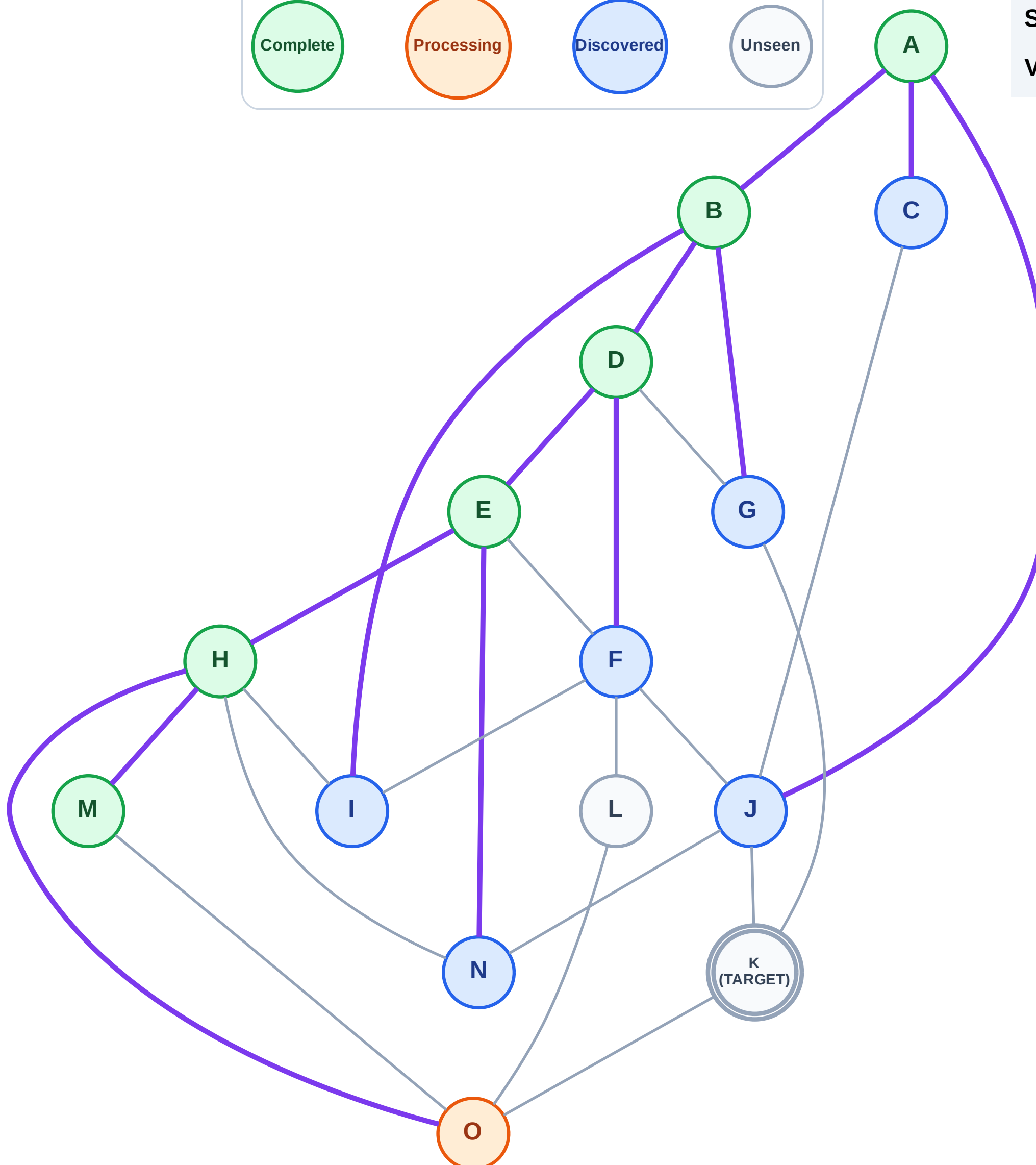
Discovered



Unseen

Stack (top at right): J | C | I | G | F | N

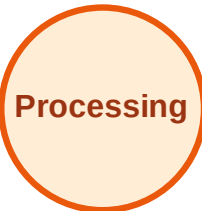
Visited: A, B, D, E, H, M



DFS Traversal

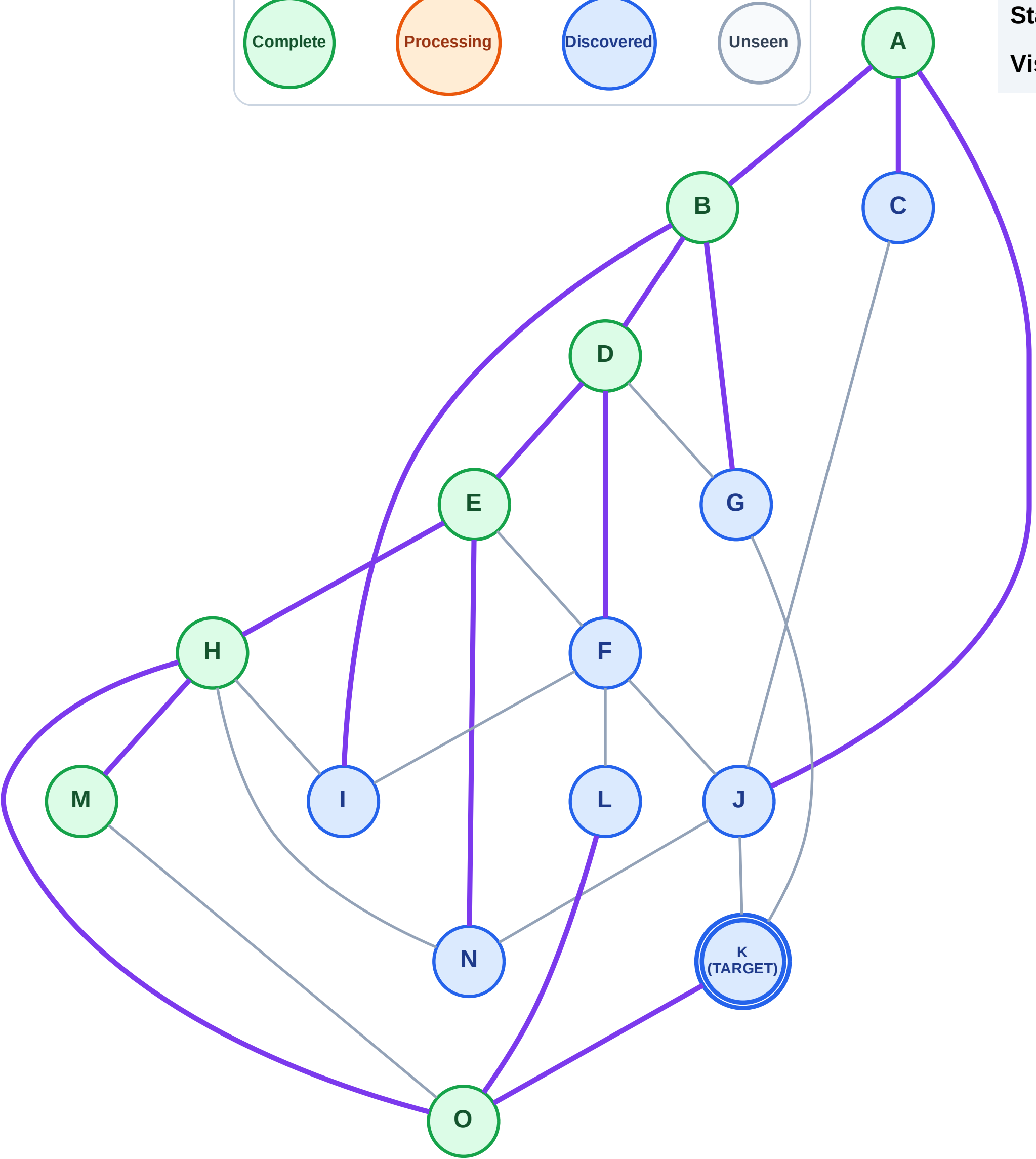
Step 15: Discover L, K from O

Legend



Stack (top at right): J | C | I | G | F | N | L | K

Visited: A, B, D, E, H, M, O



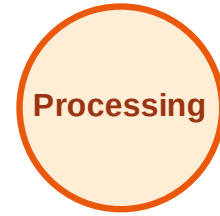
DFS Traversal

Step 16: Process node K

Legend



Complete



Processing



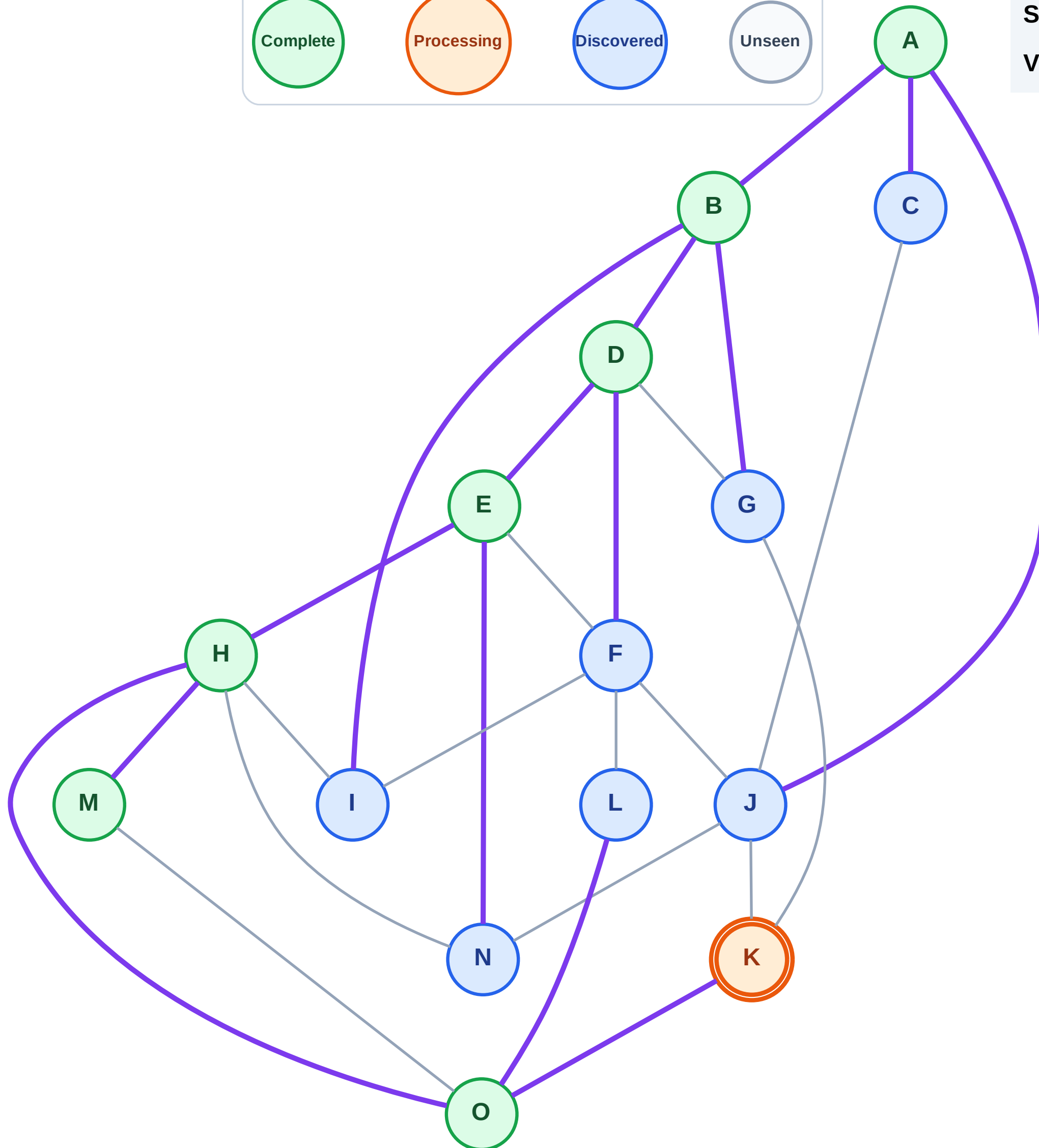
Discovered



Unseen

Stack (top at right): J | C | I | G | F | N | L

Visited: A, B, D, E, H, M, O



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (6 hops): A → B → D → E → H → O → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | I | G | F | N | L

Visited: A, B, D, E, H, K, M, O

